剑桥面试备考题选择

Cambridge Interview Preparation — Question Selections

Part I The Essentials — what is tested, and what is not

Part II Question Selections — Isaac (physics · mathematics) · UKChO (chemistry)

Five-band frequency standard (v1.3) · All question titles are the original English from the source collections

Contents

Contents 2

Part I — The Essentials 3

I. What the Interviewers Are Really Testing (all three subjects, highest weight) 3

II. Physics (the main battlefield) 3

III. Chemistry (a structural second battlefield for physics-track applicants — expect it) 3

IV. Mathematics (the tool subject) 4

V. The One-Sentence Principle 4

Part II — Question Selections 5

II-A. Isaac Physics . . . 5

II-A.1 Physics · D3 × Very High (29 questions) . . . 5

II-A.2 Physics · D2 × Very High (123 questions) . . . 6

II-A.3 Physics · D3 × High (12 questions) . . . 10

II-A.4 Physics · D2 × High (70 questions) . . . 10

II-B. Isaac Mathematics . . . 14

II-B.1 Mathematics · D3 × Very High (23 questions) . . . 14

II-B.2 Mathematics · D2 × Very High (42 questions) . . . 14

II-B.3 Mathematics · D3 × High (32 questions) . . . 16

II-B.4 Mathematics · D2 × High (3 questions) . . . 18

II-C. UKChO (Chemistry) . . . 19

II-C.1 Very High Frequency (216 items) — the interview backbone . . . 19

II-C.2 Low & Very Low Frequency (203 items) — safe to deprioritise . . . 30

Part I — The Essentials

What Cambridge NST (physics track) interviews test, and what they essentially do not.

Cambridge Natural Sciences (Physics Track) Interviews: The Essentials

I. What the Interviewers Are Really Testing (all three subjects, highest weight)

1. **Teachability** — when stuck, articulate exactly where and why, take the hint, and move on. This is the core of the marking scale; going silent is the costliest mistake;

2. **First-principles reasoning** — build the argument from conservation laws, dimensions, symmetry and limiting behaviour rather than reaching for a formula;

3. **Thinking made visible** — talk while you write, sketch, read tables. What is scored is the quality of the process, not the final answer: a perfect answer with no reasoning earns little;

4. **Estimation and dimensional sense** — order-of-magnitude estimates appear as questions in their own right, and unit consistency is an explicit marking point;

5. **Reading data and explaining anomalies** — the benchmark style is "trend plus anomaly": state the normal rule, locate the exception, and give the microscopic reason.

II. Physics (the main battlefield)

Tested

- Mechanics: force analysis, energy and momentum bookkeeping, projectile/circular/collision problems, variable mass (rockets), constrained multi-body systems;
- Estimation and dimensions; sketching functions — including non-elementary ones such as sin x/x and x^x — with a fixed routine;
- Electromagnetism: the Lorentz force, qualitative trajectories in non-uniform magnetic fields, field superposition and symmetry;
- Circuits: internal resistance, non-ideal components (diodes, LDRs), RC behaviour and power;
- The molecular picture in thermal physics, modelling with differential equations, and reasoning via limiting and boundary cases.

Essentially not tested

Quantitative relativity or quantum mechanics; unit-cell and density calculations; moments of inertia; AC impedance and transformers; astrophysical detail; competition-only techniques.

III. Chemistry (a structural second battlefield for physics-track applicants — expect it)

Tested

- Bonding and structure: discriminating bond types, VSEPR and geometry, dot-and-cross diagrams, intermolecular forces and hydrogen bonding;
- The backbone from electronic structure through periodicity to properties, including the attribution of anomalies (the F2 bond-energy anomaly is the classic);
- Physical chemistry: calorimetry and combustion enthalpy, Hess's law, bond enthalpies, $\Delta G = \Delta H - T\Delta S$, phase diagrams and colligative properties explained through entropy, acids and bases with pKa;

- A qualitative account of d–d transitions and complex colours; isomerism and spatial arrangement (axial chirality, ring strain).

Essentially not tested

Systematic organic synthesis and mechanisms (organic chemistry appears only as a vehicle for structural questions); quantitative crystal-field or molecular-orbital theory; Born–Haber cycles; unit-cell calculations; rote memory such as flame tests and characteristic precipitates; multi-step total synthesis.

IV. Mathematics (the tool subject)

Tested

Sketching functions and reading their key features; algebraic manipulation and structural recognition (symmetric expressions, recurrences); the meaning and the technique of calculus (implicit and logarithmic differentiation, integration by parts and substitution); sequences and recurrences; trigonometry; sensitivity to conditions ("what happens if this one thing changes"); rigour in proof.

Essentially not tested

Further-Maths-specific topics — complex numbers, matrices, eigenvalues, hyperbolic functions: it is enough to follow the argument and respond; ε–δ formal limits; competition-style inequality constructions.

V. The One-Sentence Principle

Drill what you already know until you can **explain it aloud**: every trend as "the normal rule → where the anomaly lies → the microscopic cause"; every estimate with its assumptions, units and dominant source of error; and, for in-person interviews, practise talking while you write on a whiteboard.

Part II — Question Selections

II-A. Isaac Physics

II-A.1 Physics · D3 × Very High (29 questions)

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Battle Bus	Physics	P1 Mechanics	Battle Bus	P1	D3	Very High
Climbing up a Corner	Physics	P1 Mechanics	Climbing up a Corner	P1	D3	Very High
Collision in Deep Space	Physics	P1 Mechanics	Collision in Deep Space	P1	D3	Very High
Crossing a River	Physics	P1 Mechanics	Crossing a River	P1	D3	Very High
Jumping Over a Log	Physics	P1 Mechanics	Jumping Over a Log	P1	D3	Very High
Maximum Deflection of a Particle	Physics	P1 Mechanics	Maximum Deflection of a Particle	P1	D3	Very High
Projectile on a String	Physics	P1 Mechanics	Projectile on a String	P1	D3	Very High
Pyramid of Spheres	Physics	P1 Mechanics	Pyramid of Spheres	P1	D3	Very High
Rod and Circle	Physics	P1 Mechanics	Rod and Circle	P1	D3	Very High
Rods, Rings and Strings	Physics	P1 Mechanics	Rods, Rings and Strings	P1	D3	Very High
Rope Between Inclines	Physics	P1 Mechanics	Rope Between Inclines	P1	D3	Very High
Superball	Physics	P1 Mechanics	Superball	P1	D3	Very High
Symmetric Hanging Chain	Physics	P1 Mechanics	Symmetric Hanging Chain	P1	D3	Very High
The Harbour Entrance	Physics	P1 Mechanics	The Harbour Entrance	P1	D3	Very High
Three Cylinders	Physics	P1 Mechanics	Three Cylinders	P1	D3	Very High
Three Spheres	Physics	P1 Mechanics	Three Spheres	P1	D3	Very High
Transfer of Energy in Collisions	Physics	P1 Mechanics	Transfer of Energy in Collisions	P1	D3	Very High
Capacitive Touchscreen	Physics	P3 Electricity & Circuits	Capacitive Touchscreen	P1	D3	Very High
Charging Two Capacitors	Physics	P3 Electricity & Circuits	Charging Two Capacitors	P1	D3	Very High
Discharging Capacitor	Physics	P3 Electricity & Circuits	Discharging Capacitor	P1	D3	Very High
Wire of Varying Area	Physics	P3 Electricity & Circuits	Wire of Varying Area	P1	D3	Very High
A Candy Cane	Physics	P1 Mechanics	A Candy Cane	P2	D3	Very High
A Hinged Rod	Physics	P1 Mechanics	A Hinged Rod	P2	D3	Very High
A Quarter Cylinder	Physics	P1 Mechanics	A Quarter Cylinder	P2	D3	Very High
A Stack of Gutters	Physics	P1 Mechanics	A Stack of Gutters	P2	D3	Very High
Flying in Circles	Physics	P1 Mechanics	Flying in Circles	P2	D3	Very High
Frictional Circular Motion	Physics	P1 Mechanics	Frictional Circular Motion	P2	D3	Very High
Loop the Loop	Physics	P1 Mechanics	Loop the Loop	P2	D3	Very High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Rising Hoop	Physics	P1 Mechanics	Rising Hoop	P2	D3	Very High

II-A.2 Physics · D2 × Very High (123 questions)

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
A Ballistic Pendulum	Physics	P1 Mechanics	A Ballistic Pendulum	P1	D2	Very High
A Firework	Physics	P1 Mechanics	A Firework	P1	D2	Very High
A Jet in the Fog	Physics	P1 Mechanics	A Jet in the Fog	P1	D2	Very High
A Power Problem	Physics	P1 Mechanics	A Power Problem	P1	D2	Very High
A Search Spiral	Physics	P1 Mechanics	A Search Spiral	P1	D2	Very High
Angle Between Two Identical Masses	Physics	P1 Mechanics	Angle Between Two Identical Masses	P1	D2	Very High
Angle of a Chip in Golf	Physics	P1 Mechanics	Angle of a Chip in Golf	P1	D2	Very High
Bounce of a Spinning Ball	Physics	P1 Mechanics	Bounce of a Spinning Ball	P1	D2	Very High
Bouncing a Ball Across a Half-Pipe	Physics	P1 Mechanics	Bouncing a Ball Across a Half-Pipe	P1	D2	Very High
Broken Cannon	Physics	P1 Mechanics	Broken Cannon	P1	D2	Very High
Building a Tennis Court	Physics	P1 Mechanics	Building a Tennis Court	P1	D2	Very High
Careless Astronauts	Physics	P1 Mechanics	Careless Astronauts	P1	D2	Very High
Changing Perspective	Physics	P1 Mechanics	Changing Perspective	P1	D2	Very High
Connected Masses	Physics	P1 Mechanics	Connected Masses	P1	D2	Very High
Conservation of Momentum 4	Physics	P1 Mechanics	Conservation of Momentum 4	P1	D2	Very High
Conservation of Momentum 5	Physics	P1 Mechanics	Conservation of Momentum 5	P1	D2	Very High
Conservation of Momentum 6	Physics	P1 Mechanics	Conservation of Momentum 6	P1	D2	Very High
Cosmic Reflection	Physics	P1 Mechanics	Cosmic Reflection	P1	D2	Very High
Cycling Through Wind	Physics	P1 Mechanics	Cycling Through Wind	P1	D2	Very High
Dropping a Mass on a Windmill	Physics	P1 Mechanics	Dropping a Mass on a Windmill	P1	D2	Very High
Falling Chain	Physics	P1 Mechanics	Falling Chain	P1	D2	Very High
Festival Banner	Physics	P1 Mechanics	Festival Banner	P1	D2	Very High
Finger Balance	Physics	P1 Mechanics	Finger Balance	P1	D2	Very High
Force of a Bouncing Ball	Physics	P1 Mechanics	Force of a Bouncing Ball	P1	D2	Very High
Geometry and Inverse Square Laws	Physics	P1 Mechanics	Geometry and Inverse Square Laws	P1	D2	Very High
Gravtube	Physics	P1 Mechanics	Gravtube	P1	D2	Very High
Hot Air Balloon Ropes	Physics	P1 Mechanics	Hot Air Balloon Ropes	P1	D2	Very High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Impulse and an Inclined Plane	Physics	P1 Mechanics	Impulse and an Inclined Plane	P1	D2	Very High
Inelastic and Elastic Collisions	Physics	P1 Mechanics	Inelastic and Elastic Collisions	P1	D2	Very High
Lifting a Rod	Physics	P1 Mechanics	Lifting a Rod	P1	D2	Very High
Longest Trajectory of a Thrown Ball	Physics	P1 Mechanics	Longest Trajectory of a Thrown Ball	P1	D2	Very High
Love Bugs	Physics	P1 Mechanics	Love Bugs	P1	D2	Very High
Maximum Angle Throw	Physics	P1 Mechanics	Maximum Angle Throw	P1	D2	Very High
Modelling Earth's Gravity	Physics	P1 Mechanics	Modelling Earth's Gravity	P1	D2	Very High
Newtonian Gravity 8	Physics	P1 Mechanics	Newtonian Gravity 8	P1	D2	Very High
Oblique Collisions and Walls	Physics	P1 Mechanics	Oblique Collisions and Walls	P1	D2	Very High
Pedal Power	Physics	P1 Mechanics	human power	P1	D2	Very High
Picking up a Chain	Physics	P1 Mechanics	Picking up a Chain	P1	D2	Very High
Pop-up Toy	Physics	P1 Mechanics	Pop-up Toy	P1	D2	Very High
Pulleys on a Table	Physics	P1 Mechanics	Pulleys on a Table	P1	D2	Very High
Reaching a Sloping Roof	Physics	P1 Mechanics	Reaching a Sloping Roof	P1	D2	Very High
Restitution: Sphere Collision 2	Physics	P1 Mechanics	Restitution: Sphere Collision 2	P1	D2	Very High
Restitution and a Bouncing Ball	Physics	P1 Mechanics	Restitution and a Bouncing Ball	P1	D2	Very High
Restitution: Sphere Collision	Physics	P1 Mechanics	Restitution: Sphere Collision	P1	D2	Very High
Road Collision	Physics	P1 Mechanics	Road Collision	P1	D2	Very High
Rotational Tug of War	Physics	P1 Mechanics	Rotational Tug of War	P1	D2	Very High
Shooting a Balloon	Physics	P1 Mechanics	Shooting a Balloon	P1	D2	Very High
Size of a Combined Force	Physics	P1 Mechanics	Size of a Combined Force	P1	D2	Very High
Smoothness Test	Physics	P1 Mechanics	Smoothness Test	P1	D2	Very High
Snowflake Fall Speeds	Physics	P1 Mechanics	Snowflake Fall Speeds	P1	D2	Very High
Star Potential Energy	Physics	P1 Mechanics	Star Potential Energy	P1	D2	Very High
Stopping Distances	Physics	P1 Mechanics	Stopping Distances	P1	D2	Very High
Stuck in the Mud	Physics	P1 Mechanics	Stuck in the Mud	P1	D2	Very High
Swimming to a Boat	Physics	P1 Mechanics	Swimming to a Boat	P1	D2	Very High
Test Rocket	Physics	P1 Mechanics	Test Rocket	P1	D2	Very High
The Ant and the Tortoise	Physics	P1 Mechanics	The Ant and the Tortoise	P1	D2	Very High
The Basketball Player	Physics	P1 Mechanics	The Basketball Player	P1	D2	Very High
The Bolt Thrower	Physics	P1 Mechanics	The Bolt Thrower	P1	D2	Very High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
The Bouncing Ball	Physics	P1 Mechanics	The Bouncing Ball	P1	D2	Very High
The Hose Pipe	Physics	P1 Mechanics	The Hose Pipe	P1	D2	Very High
The Lifeguard	Physics	P1 Mechanics	The Lifeguard	P1	D2	Very High
The Water Wheel	Physics	P1 Mechanics	The Water Wheel	P1	D2	Very High
The Weighing Machine	Physics	P1 Mechanics	The Weighing Machine	P1	D2	Very High
Throwing a Ball up a Flat Slope	Physics	P1 Mechanics	Throwing a Ball up a Flat Slope	P1	D2	Very High
Thrown off a Cliff	Physics	P1 Mechanics	Thrown off a Cliff	P1	D2	Very High
Towel on a Washing Line	Physics	P1 Mechanics	Towel on a Washing Line	P1	D2	Very High
Trajectories 8	Physics	P1 Mechanics	Trajectories 8	P1	D2	Very High
Trajectories 9	Physics	P1 Mechanics	Trajectories 9	P1	D2	Very High
Two Spinning Balls	Physics	P1 Mechanics	Two Spinning Balls	P1	D2	Very High
Wind Turbine Power	Physics	P1 Mechanics	Wind Turbine Power	P1	D2	Very High
Work Done by a Wheel	Physics	P1 Mechanics	Work Done by a Wheel	P1	D2	Very High
A Capacitive Conundrum	Physics	P3 Electricity & Circuits	A Capacitive Conundrum	P1	D2	Very High
A Particle Current	Physics	P3 Electricity & Circuits	A Particle Current	P1	D2	Very High
A Resistive Shape	Physics	P3 Electricity & Circuits	A Resistive Shape	P1	D2	Very High
Capacitor Energy Bank	Physics	P3 Electricity & Circuits	Capacitor Energy Bank	P1	D2	Very High
Charge Carriers 6	Physics	P3 Electricity & Circuits	Charge Carriers 6	P1	D2	Very High
Charge Carriers II 6	Physics	P3 Electricity & Circuits	Charge Carriers II 6	P1	D2	Very High
Conveying Current	Physics	P3 Electricity & Circuits	Conveying Current	P1	D2	Very High
Current Between Resistors	Physics	P3 Electricity & Circuits	Current Between Resistors	P1	D2	Very High
Different Spring Circuit	Physics	P3 Electricity & Circuits	Different Spring Circuit	P1	D2	Very High
Drift Velocity	Physics	P3 Electricity & Circuits	Drift Velocity	P1	D2	Very High
Fermi Velocity	Physics	P3 Electricity & Circuits	Fermi Velocity	P1	D2	Very High
Measuring Resistances	Physics	P3 Electricity & Circuits	Measuring Resistances	P1	D2	Very High
Microscopic Current	Physics	P3 Electricity & Circuits	Microscopic Current	P1	D2	Very High
Parallel Plates	Physics	P3 Electricity & Circuits	Parallel Plates	P1	D2	Very High
Powering Up	Physics	P3 Electricity & Circuits	Powering Up	P1	D2	Very High
Similar Spring Circuit	Physics	P3 Electricity & Circuits	Similar Spring Circuit	P1	D2	Very High
Variable Capacitor	Physics	P3 Electricity & Circuits	Variable Capacitor	P1	D2	Very High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Where do the Charges Go?	Physics	P3 Electricity & Circuits	Where do the Charges Go?	P1	D2	Very High
Time to Reach a Star	Physics	P9 Units, Dimensions & Estimation	Time to Reach a Star	P1	D2	Very High
A Chain on a Peg	Physics	P1 Mechanics	A Chain on a Peg	P2	D2	Very High
A Coiled Rope	Physics	P1 Mechanics	A Coiled Rope	P2	D2	Very High
A Fairground Ride	Physics	P1 Mechanics	A Fairground Ride	P2	D2	Very High
A Particle in a Bowl	Physics	P1 Mechanics	A Particle in a Bowl	P2	D2	Very High
A Uniform Ladder	Physics	P1 Mechanics	A Uniform Ladder	P2	D2	Very High
Banked Tracks for Turning 4	Physics	P1 Mechanics	Banked Tracks for Turning 4	P2	D2	Very High
Banked Tracks for Turning 5	Physics	P1 Mechanics	Banked Tracks for Turning 5	P2	D2	Very High
Banked Tracks for Turning 6	Physics	P1 Mechanics	Banked Tracks for Turning 6	P2	D2	Very High
Banked Tracks for Turning 8	Physics	P1 Mechanics	Banked Tracks for Turning 8	P2	D2	Very High
Banked Tracks for Turning 9	Physics	P1 Mechanics	Banked Tracks for Turning 9	P2	D2	Very High
Car on a Roundabout	Physics	P1 Mechanics	Car on a Roundabout	P2	D2	Very High
Centre of Mass: Prism Topple	Physics	P1 Mechanics	Centre of Mass: Prism Topple	P2	D2	Very High
Centre of Mass: Prism Topple 2	Physics	P1 Mechanics	Centre of Mass: Prism Topple 2	P2	D2	Very High
Centripetal Acceleration 5	Physics	P1 Mechanics	Centripetal Acceleration 5	P2	D2	Very High
Centripetal Acceleration 6	Physics	P1 Mechanics	Centripetal Acceleration 6	P2	D2	Very High
Conical Pendulum 4	Physics	P1 Mechanics	Conical Pendulum 4	P2	D2	Very High
Conical Pendulum 5	Physics	P1 Mechanics	Conical Pendulum 5	P2	D2	Very High
Conical Pendulum 7	Physics	P1 Mechanics	Conical Pendulum 7	P2	D2	Very High
Cornering on a Smooth Surface	Physics	P1 Mechanics	Cornering on a Smooth Surface	P2	D2	Very High
Double Decker Bus	Physics	P1 Mechanics	Double Decker Bus	P2	D2	Very High
Elastic Circular Motion	Physics	P1 Mechanics	Elastic Circular Motion	P2	D2	Very High
Hammer Throwing	Physics	P1 Mechanics	Hammer Throwing	P2	D2	Very High
Hanging a Non-uniform Bar	Physics	P1 Mechanics	Hanging a Non-uniform Bar	P2	D2	Very High
Leaning Ladder	Physics	P1 Mechanics	Leaning Ladder	P2	D2	Very High
Moments 6	Physics	P1 Mechanics	Moments 6	P2	D2	Very High
Moments 7	Physics	P1 Mechanics	Moments 7	P2	D2	Very High
Moving in a Circle 6	Physics	P1 Mechanics	Moving in a Circle 6	P2	D2	Very High
Moving in a Circle 7	Physics	P1 Mechanics	Moving in a Circle 7	P2	D2	Very High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Shelf and Brackets	Physics	P1 Mechanics	Shelf and Brackets	P2	D2	Very High
Sliding Inside a Hoop	Physics	P1 Mechanics	Sliding Inside a Hoop	P2	D2	Very High
Tipping a Tumbler	Physics	P1 Mechanics	Tipping a Tumbler	P2	D2	Very High
Toppling Block	Physics	P1 Mechanics	Toppling Block	P2	D2	Very High
Trapdoor	Physics	P1 Mechanics	Trapdoor	P2	D2	Very High

II-A.3 Physics · D3 × High (12 questions)

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Mercury in a U-Tube	Physics	P4 Oscillations & Waves	Mercury in a U-Tube	P1	D3	High
Spring Pick-up	Physics	P4 Oscillations & Waves	Spring Pick-up	P1	D3	High
Heating a Mixture	Physics	P6 Thermal Physics	Heating a Mixture	P1	D3	High
Aiming an Oscilloscope Beam	Physics	P2 Fields & Potential	Aiming an Oscilloscope Beam	P2	D3	High
Electric Projectile	Physics	P2 Fields & Potential	Electric Projectile	P2	D3	High
Force From Electric Dipole	Physics	P2 Fields & Potential	Force From Electric Dipole	P2	D3	High
Inkjet Printing	Physics	P2 Fields & Potential	Inkjet Printing	P2	D3	High
Capacitor & Diode Rectifier	Physics	P3 Electricity & Circuits	Capacitor & Diode Rectifier	P2	D3	High
Capacitors and Resistors 7	Physics	P3 Electricity & Circuits	Capacitors and Resistors 7	P2	D3	High
Potentiometer Circuit With Capacitor	Physics	P3 Electricity & Circuits	Potentiometer Circuit With Capacitor	P2	D3	High
Electromagnetic Induction in a Disc	Physics	P10 Electromagnetism	Electromagnetic Induction in a Disc	P2	D3	High
Slowing Down A Bomb	Physics	P10 Electromagnetism	Slowing Down A Bomb	P2	D3	High

II-A.4 Physics · D2 × High (70 questions)

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
A Bobbing Buoy	Physics	P4 Oscillations & Waves	A Bobbing Buoy	P1	D2	High
A Diatomic Molecule	Physics	P4 Oscillations & Waves	A Diatomic Molecule	P1	D2	High
A Mass, a Rod and a Spring	Physics	P4 Oscillations & Waves	A Mass, a Rod and a Spring	P1	D2	High
An Anchored Buoy	Physics	P4 Oscillations & Waves	An Anchored Buoy	P1	D2	High
Ball Dropped Onto Spring	Physics	P4 Oscillations & Waves	Ball Dropped Onto Spring	P1	D2	High
Capacitor on Hooke's	Physics	P4 Oscillations & Waves	Capacitor on Hooke's	P1	D2	High
Car Suspension Over Bumps	Physics	P4 Oscillations & Waves	Car Suspension Over Bumps	P1	D2	High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Chopping up Springs	Physics	P4 Oscillations & Waves	Chopping up Springs	P1	D2	High
Circular Pendulum	Physics	P4 Oscillations & Waves	Circular Pendulum	P1	D2	High
Conical Pendulum	Physics	P4 Oscillations & Waves	Conical Pendulum	P1	D2	High
Ionic Molecule	Physics	P4 Oscillations & Waves	Ionic Molecule	P1	D2	High
Mass Resting on a Spring	Physics	P4 Oscillations & Waves	Mass Resting on a Spring	P1	D2	High
Oscillators 7	Physics	P4 Oscillations & Waves	Oscillators 7	P1	D2	High
Rope Swing	Physics	P4 Oscillations & Waves	Rope Swing	P1	D2	High
SHM of a Simple Scale	Physics	P4 Oscillations & Waves	SHM of a Simple Scale	P1	D2	High
SHM of a Wire Loop	Physics	P4 Oscillations & Waves	SHM of a Wire Loop	P1	D2	High
SHM Perpendicular to a Spring	Physics	P4 Oscillations & Waves	SHM Perpendicular to a Spring	P1	D2	High
Simple Pendulum 2	Physics	P4 Oscillations & Waves	Simple Pendulum 2	P1	D2	High
Simple Pendulum 4	Physics	P4 Oscillations & Waves	Simple Pendulum 4	P1	D2	High
Spring Triangle	Physics	P4 Oscillations & Waves	Spring Triangle	P1	D2	High
Tension in a Simple Pendulum	Physics	P4 Oscillations & Waves	Tension in a Simple Pendulum	P1	D2	High
Weighing at Sea	Physics	P4 Oscillations & Waves	Weighing at Sea	P1	D2	High
Air Changes	Physics	P6 Thermal Physics	Air Changes	P1	D2	High
Barometric Formula	Physics	P6 Thermal Physics	Barometric Formula	P1	D2	High
Energy of a Bullet	Physics	P6 Thermal Physics	Energy of a Bullet	P1	D2	High
Gas Laws 6	Physics	P6 Thermal Physics	Gas Laws 6	P1	D2	High
Gas Laws 7	Physics	P6 Thermal Physics	Gas Laws 7	P1	D2	High
Heating a Wire	Physics	P6 Thermal Physics	Heating a Wire	P1	D2	High
Ice Cream Cone	Physics	P6 Thermal Physics	Ice Cream Cone	P1	D2	High
Latent Heat and Heat Capacity 5	Physics	P6 Thermal Physics	Latent Heat and Heat Capacity 5	P1	D2	High
Melting a Snowman	Physics	P6 Thermal Physics	Melting a Snowman	P1	D2	High
Sea Level Rise	Physics	P6 Thermal Physics	Sea Level Rise	P1	D2	High
Brownian Motion	Physics	P6 Thermal Physics	Brownian Motion	P1	D2	High
A Rod in a Non-Uniform Field	Physics	P2 Fields & Potential	A Rod in a Non-Uniform Field	P2	D2	High
Accelerating Particles	Physics	P2 Fields & Potential	Accelerating Particles	P2	D2	High
Aligning Dipoles	Physics	P2 Fields & Potential	Aligning Dipoles	P2	D2	High
Between Two Masses	Physics	P2 Fields & Potential	Between Two Masses	P2	D2	High
Charged Stars	Physics	P2 Fields & Potential	Charged Stars	P2	D2	High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Drops of Oil	Physics	P2 Fields & Potential	Drops of Oil	P2	D2	High
Electric Charge of Earth	Physics	P2 Fields & Potential	Electric Charge of Earth	P2	D2	High
Electric Deflection 7	Physics	P2 Fields & Potential	Electric Deflection 7	P2	D2	High
Electric Deflection 8	Physics	P2 Fields & Potential	Electric Deflection 8	P2	D2	High
Electric Field Near Point Charges 5	Physics	P2 Fields & Potential	Electric Field Near Point Charges 5	P2	D2	High
Electric Field Near Point Charges 6	Physics	P2 Fields & Potential	Electric Field Near Point Charges 6	P2	D2	High
Electric Levitation	Physics	P2 Fields & Potential	Electric Levitation	P2	D2	High
Electric Shapes	Physics	P2 Fields & Potential	Electric Shapes	P2	D2	High
Electrostatic Shield	Physics	P2 Fields & Potential	Electrostatic Shield	P2	D2	High
Energy and Potential in Electric Fields 7	Physics	P2 Fields & Potential	Energy and Potential in Electric Fields 7	P2	D2	High
Liquid Pool	Physics	P2 Fields & Potential	Liquid Pool	P2	D2	High
Oil Drop Experiment	Physics	P2 Fields & Potential	Oil Drop Experiment	P2	D2	High
Vectors in Fields	Physics	P2 Fields & Potential	Vectors in Fields	P2	D2	High
An Oscillating Capacitor	Physics	P3 Electricity & Circuits	An Oscillating Capacitor	P2	D2	High
Conical Wavefront	Physics	P4 Oscillations & Waves	Conical Wavefront	P2	D2	High
Decibels	Physics	P4 Oscillations & Waves	Decibels	P2	D2	High
Cooling Spheres	Physics	P6 Thermal Physics	Cooling Spheres	P2	D2	High
Aeroplane Potential Difference	Physics	P10 Electromagnetism	Aeroplane Potential Difference	P2	D2	High
Charging a Falling Sphere	Physics	P10 Electromagnetism	Charging a Falling Sphere	P2	D2	High
Circular Paths in Magnetic Fields 3	Physics	P10 Electromagnetism	Circular Paths in Magnetic Fields 3	P2	D2	High
Circular Paths in Magnetic Fields 4	Physics	P10 Electromagnetism	Circular Paths in Magnetic Fields 4	P2	D2	High
Conductor in a Magnetic Field 5	Physics	P10 Electromagnetism	Conductor in a Magnetic Field 5	P2	D2	High
Conductor in a Magnetic Field 6	Physics	P10 Electromagnetism	Conductor in a Magnetic Field 6	P2	D2	High
Induction in a Moving Wire 8	Physics	P10 Electromagnetism	Induction in a Moving Wire 8	P2	D2	High
Interaction of a Ring and a Coil	Physics	P10 Electromagnetism	Interaction of a Ring and a Coil	P2	D2	High
Magnetic Flux and Faraday's Law 7	Physics	P10 Electromagnetism	Magnetic Flux and Faraday's Law 7	P2	D2	High
Particle in a Magnetic Field 3	Physics	P10 Electromagnetism	Particle in a Magnetic Field 3	P2	D2	High
Ring Dropped Into Magnet	Physics	P10 Electromagnetism	Ring Dropped Into Magnet	P2	D2	High
Sliding Wire	Physics	P10 Electromagnetism	Sliding Wire	P2	D2	High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
The Catapult Effect	Physics	P10 Electromagnetism	The Catapult Effect	P2	D2	High
Charges in Magnetic Fields 1	Physics	P10 Electromagnetism	Charges in Magnetic Fields 1	P2	D2	High
Charges in Magnetic Fields 2	Physics	P10 Electromagnetism	Charges in Magnetic Fields 2	P2	D2	High

II-B. Isaac Mathematics

II-B.1 Mathematics · D3 × Very High (23 questions)

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Graph Interpreting: Beats	Mathematics	M2 Trigonometry & Geometry	beat-frequency graphs	P1	D3	Very High
A Vertical Throw	Mathematics	M3 Calculus	vertical throw (integration)	P1	D3	Very High
Population Growth	Mathematics	M3 Calculus	population growth (exponential)	P1	D3	Very High
Population Near Equilibrium	Mathematics	M3 Calculus	stability near equilibrium	P1	D3	Very High
Simple Harmonic Motion 3	Mathematics	M3 Calculus	SHM (variant)	P1	D3	Very High
Sketching Oscillator Amplitude	Mathematics	M3 Calculus	Sketching Oscillator Amplitude	P1	D3	Very High
Sketching a Lorentzian Function	Mathematics	M3 Calculus	Sketching a Lorentzian Function	P1	D3	Very High
Sketching a Rational Function 3	Mathematics	M3 Calculus	rational functions sketch	P1	D3	Very High
Partial Fractions - Cubed Linear Factor	Mathematics	M1 Algebra & Functions	Partial Fractions - Cubed Linear Factor	P2	D3	Very High
Partial Fractions Applied to Other Functions	Mathematics	M1 Algebra & Functions	partial fractions of application	P2	D3	Very High
Blackbody Radiation	Mathematics	M3 Calculus	Blackbody Radiation	P2	D3	Very High
Geometry of Log Horns	Mathematics	M3 Calculus	Geometry of Log Horns	P2	D3	Very High
Integrating 1/x	Mathematics	M3 Calculus	Integrating 1/x	P2	D3	Very High
Integrating Factors 4	Mathematics	M3 Calculus	Integrating Factors 4	P2	D3	Very High
Integration Using Trig Identities 2	Mathematics	M3 Calculus	Integration Using Trig Identities 2	P2	D3	Very High
Integration Using Trig Identities 3	Mathematics	M3 Calculus	Integration Using Trig Identities 3	P2	D3	Very High
Integration with Partial Fractions 3	Mathematics	M3 Calculus	Integration with Partial Fractions 3	P2	D3	Very High
Log Horns: Area & Volume	Mathematics	M3 Calculus	Log Horns: Area & Volume	P2	D3	Very High
Parametric Volumes 1	Mathematics	M3 Calculus	Parametric Volumes 1	P2	D3	Very High
Parametric Volumes 2	Mathematics	M3 Calculus	Parametric Volumes 2	P2	D3	Very High
Parametric Volumes 3	Mathematics	M3 Calculus	Parametric Volumes 3	P2	D3	Very High
Stationary Points 4	Mathematics	M3 Calculus	Stationary Points 4	P2	D3	Very High
Stationary Points 5	Mathematics	M3 Calculus	Stationary Points 5	P2	D3	Very High

II-B.2 Mathematics · D2 × Very High (42 questions)

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Apparent Magnitudes	Mathematics	M1 Algebra & Functions	Apparent Magnitudes	P1	D2	Very High
Factorising Quadratic Equations 11	Mathematics	M1 Algebra & Functions	Factorising Quadratic Equations 11	P1	D2	Very High
Reflection and Symmetry	Mathematics	M1 Algebra & Functions	Reflection and Symmetry	P1	D2	Very High
Working With Terms	Mathematics	M1 Algebra & Functions	Working With Terms	P1	D2	Very High
Area Under a Curve	Mathematics	M3 Calculus	Area Under a Curve	P1	D2	Very High
Graph Interpreting: Exponentials and Logs	Mathematics	M3 Calculus	exponential / logarithmic graph interpretation	P1	D2	Very High
Graph Interpreting: Product of Functions 1	Mathematics	M3 Calculus	Graph Interpreting: Product of Functions 1	P1	D2	Very High
Graph Interpreting: Quantum Harmonic Oscillator	Mathematics	M3 Calculus	Graph Interpreting: Quantum Harmonic Oscillator	P1	D2	Very High
Graph Interpreting: Rational Functions 1	Mathematics	M3 Calculus	rational functions graphs	P1	D2	Very High
Graph Interpreting: Rational Functions 2	Mathematics	M3 Calculus	rational functions graphs (variant)	P1	D2	Very High
Graph Interpreting: Reciprocal Function	Mathematics	M3 Calculus	reciprocal functions graphs	P1	D2	Very High
Graph Interpreting: sin Squared	Mathematics	M3 Calculus	sin2 graphs	P1	D2	Very High
Graph Interpreting: Sum of sin Functions	Mathematics	M3 Calculus	Graph Interpreting: Sum of sin Functions	P1	D2	Very High
Graph Interpreting: Sums of Functions 2	Mathematics	M3 Calculus	Graph Interpreting: Sums of Functions 2	P1	D2	Very High
Sketching a Cosine-Gaussian Graph	Mathematics	M3 Calculus	Sketching a Cosine-Gaussian Graph	P1	D2	Very High
Sketching a Cubic Modulus Function	Mathematics	M3 Calculus	Sketching a Cubic Modulus Function	P1	D2	Very High
Sketching a Modulus Function Graph	Mathematics	M3 Calculus	modulus functions graphs	P1	D2	Very High
Sketching a Quadratic-Exponential Graph	Mathematics	M3 Calculus	Sketching a Quadratic-Exponential Graph	P1	D2	Very High
Sketching a Rational Function 1	Mathematics	M3 Calculus	rational functions sketch	P1	D2	Very High
Sketching a Rational Function 2	Mathematics	M3 Calculus	rational functions sketch (variant)	P1	D2	Very High
Sketching a sinc Function	Mathematics	M3 Calculus	Sketching a sinc Function	P1	D2	Very High
Sketching Fermi-Dirac Distribution	Mathematics	M3 Calculus	Sketching Fermi-Dirac Distribution	P1	D2	Very High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Sketching Graph of Gravitational Potential	Mathematics	M3 Calculus	Sketching Graph of Gravitational Potential	P1	D2	Very High
Sketching Graph of Heavily Damped System	Mathematics	M3 Calculus	Sketching Graph of Heavily Damped System	P1	D2	Very High
Sketching Graph of Motion in Viscous Medium	Mathematics	M3 Calculus	Sketching Graph of Motion in Viscous Medium	P1	D2	Very High
Sketching Graph of Rocket Speed	Mathematics	M3 Calculus	Sketching Graph of Rocket Speed	P1	D2	Very High
Sketching Graphs of Exponentials and Logs	Mathematics	M3 Calculus	exponential and logarithmic sketch	P1	D2	Very High
Sketching Graphs of Reciprocals 3	Mathematics	M3 Calculus	reciprocal functions sketch	P1	D2	Very High
Sketching Graphs of sin and tan	Mathematics	M3 Calculus	sin/tan sketch	P1	D2	Very High
Sketching Oscillator Properties	Mathematics	M3 Calculus	Sketching Oscillator Properties	P1	D2	Very High
Sketching Travel Time of Reflected Wave	Mathematics	M3 Calculus	Sketching Travel Time of Reflected Wave	P1	D2	Very High
Bohr Radius	Mathematics	M4 Estimation & Ratios	Bohr Radius	P1	D2	Very High
Cosmic Rays	Mathematics	M4 Estimation & Ratios	Cosmic Rays	P1	D2	Very High
Error in Power Dissipation	Mathematics	M4 Estimation & Ratios	Error in Power Dissipation	P1	D2	Very High
Light Pulse on a Train	Mathematics	M4 Estimation & Ratios	Light Pulse on a Train	P1	D2	Very High
Santa Flying Stunt	Mathematics	M4 Estimation & Ratios	Santa Flying Stunt	P1	D2	Very High
The Half Hour	Mathematics	M4 Estimation & Ratios	time estimation (half hour)	P1	D2	Very High
Calculus When Working in Degrees	Mathematics	M3 Calculus	Calculus When Working in Degrees	P2	D2	Very High
Maximum Energy Transfer	Mathematics	M3 Calculus	Maximum Energy Transfer	P2	D2	Very High
Maximum Power Dissipation	Mathematics	M3 Calculus	Maximum Power Dissipation	P2	D2	Very High
Stationary Points 3	Mathematics	M3 Calculus	Stationary Points 3	P2	D2	Very High
Graph Interpreting: Hyperbolic	Mathematics	M9 Further Mathematics	Graph Interpreting: Hyperbolic	P4	D2	Very High

II-B.3 Mathematics · D3 × High (32 questions)

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Circular Motion	Mathematics	M2 Trigonometry & Geometry	Circular Motion	P1	D3	High
Forces on an Aerofoil	Mathematics	M2 Trigonometry & Geometry	Forces on an Aerofoil	P1	D3	High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Sand Yachts and Ice Boats	Mathematics	M2 Trigonometry & Geometry	Sand Yachts and Ice Boats	P1	D3	High
Speed of Water Yachts	Mathematics	M2 Trigonometry & Geometry	Speed of Water Yachts	P1	D3	High
The Apparent Wind	Mathematics	M2 Trigonometry & Geometry	The Apparent Wind	P1	D3	High
The Logarithmic Spiral	Mathematics	M2 Trigonometry & Geometry	The Logarithmic Spiral	P1	D3	High
The Maximum Speed	Mathematics	M2 Trigonometry & Geometry	The Maximum Speed	P1	D3	High
Wave Interference	Mathematics	M2 Trigonometry & Geometry	Wave Interference	P1	D3	High
A Leaky Tank	Mathematics	M3 Calculus	A Leaky Tank	P1	D3	High
A Rocket	Mathematics	M3 Calculus	A Rocket	P1	D3	High
An Exponential Differential Equation	Mathematics	M3 Calculus	An Exponential Differential Equation	P1	D3	High
Discharging a Capacitor	Mathematics	M3 Calculus	Discharging a Capacitor	P1	D3	High
Discharging a Metal Sphere	Mathematics	M3 Calculus	Discharging a Metal Sphere	P1	D3	High
Falling Through Air	Mathematics	M3 Calculus	Falling Through Air	P1	D3	High
Introducing a Predator	Mathematics	M3 Calculus	Introducing a Predator	P1	D3	High
Mass on a Spring	Mathematics	M3 Calculus	Mass on a Spring	P1	D3	High
(Isaac) Newton's Law of Cooling	Mathematics	M3 Calculus	(Isaac) Newton's Law of Cooling	P1	D3	High
People	Mathematics	M3 Calculus	People	P1	D3	High
Sedimentation	Mathematics	M3 Calculus	Sedimentation	P1	D3	High
Simple Harmonic Motion 2	Mathematics	M3 Calculus	Simple Harmonic Motion 2	P1	D3	High
Solving the Logistic Equation	Mathematics	M3 Calculus	Solving the Logistic Equation	P1	D3	High
The "Universal" Equation	Mathematics	M3 Calculus	The "Universal" Equation	P1	D3	High
The Predator-Prey Equations	Mathematics	M3 Calculus	The Predator-Prey Equations	P1	D3	High
Chain Sliding off a Rough Table	Mathematics	M3 Calculus	Chain Sliding off a Rough Table	P2	D3	High
Driven Oscillator	Mathematics	M3 Calculus	Driven Oscillator	P2	D3	High
Gravitational Orbits	Mathematics	M3 Calculus	Gravitational Orbits	P2	D3	High
Quadratic Potential Well	Mathematics	M3 Calculus	Quadratic Potential Well	P2	D3	High
RLC Circuit	Mathematics	M3 Calculus	RLC Circuit	P2	D3	High
Solution by Substitution 1	Mathematics	M3 Calculus	Solution by Substitution 1	P2	D3	High

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Solution by Substitution 2	Mathematics	M3 Calculus	Solution by Substitution 2	P2	D3	High
Square Quantum Well	Mathematics	M3 Calculus	Square Quantum Well	P2	D3	High
Complex Form Solution	Mathematics	M8 Extensions	Complex Form Solution	P3	D3	High

II-B.4 Mathematics · D2 × High (3 questions)

Question Title	Subject	Module	Topic	Tier	Difficulty	Interview Frequency
Area of Isosceles Triangle	Mathematics	M2 Trigonometry & Geometry	Area of Isosceles Triangle	P1	D2	High
Plane Waves	Mathematics	M2 Trigonometry & Geometry	Plane Waves	P1	D2	High
Oscillating Disk	Mathematics	M3 Calculus	Oscillating Disk	P1	D2	High

II-C. UKChO (Chemistry)

II-C.1 Very High Frequency (216 items) — the interview backbone

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2026 Q1	Pee in the pool	(c)(ii) Shape of ClO3-	(c)(ii) Shape of ClO3-	C2 Bonding & Structure	P1	Very High
2026 Q1	Pee in the pool	(d) Urea hydrolysis	(d) Urea hydrolysis	C8 Organic Chemistry	P2	Very High
2026 Q1	Pee in the pool	(e) Dot-and-cross diagram of NH2Cl	(e) Dot-and-cross diagram of NH2Cl	C2 Bonding & Structure	P1	Very High
2026 Q1	Pee in the pool	(f) Henry's-law partial pressure of NCl3	(f) Henry's-law partial pressure of NCl3	C6 Equilibrium & Acids–Bases	P2	Very High
2026 Q2	Isocyanides	(a) Resonance structure of the isocyanide	(a) Resonance structure of the isocyanide	C8 Organic Chemistry	P2	Very High
2026 Q2	Isocyanides	(b)(i) Compound B from NaCN + allyl bromide; (ii) precipitate	(b)(i) Compound B from NaCN + allyl bromide; (ii) precipitate	C8 Organic Chemistry	P2	Very High
2026 Q2	Isocyanides	(d) Compound C and compound D	(d) Compound C and compound D	C8 Organic Chemistry	P2	Very High
2026 Q2	Isocyanides	(h) Isobel's Passerini product count	(h) Isobel's Passerini product count	C8 Organic Chemistry	P2	Very High
2026 Q2	Isocyanides	(i) Atom with stereogenic potential	(i) Atom with stereogenic potential	C8 Organic Chemistry	P2	Very High
2026 Q2	Isocyanides	(j) Cynthia's product count (each enantiomer unique)	(j) Cynthia's product count (each enantiomer unique)	C8 Organic Chemistry	P2	Very High
2026 Q2	Isocyanides	(k) Galvin's product count	(k) Galvin's product count	C8 Organic Chemistry	P2	Very High
2026 Q3	Compounds in henna tattoos	(c) Which aldehydes/ketones can form tautomers	(c) Which aldehydes/ketones can form tautomers	C8 Organic Chemistry	P2	Very High
2026 Q3	Compounds in henna tattoos	(d)(i) The two enol isomers of lawsone; (ii) the hydrogen bond	(d)(i) The two enol isomers of lawsone; (ii) the hydrogen bond	C2 Bonding & Structure	P1	Very High
2026 Q3	Compounds in henna tattoos	(f) Tautomers	(f) Tautomers	C8 Organic Chemistry	P2	Very High
2026 Q3	Compounds in henna tattoos	(g) All tautomers	(g) All tautomers	C8 Organic Chemistry	P2	Very High
2026 Q3	Compounds in henna tattoos	(h) ΔrH°(keto→enol) from bond enthalpies	(h) ΔrH°(keto→enol) from bond enthalpies	C4 Thermodynamics	P1	Very High
2026 Q3	Compounds in henna tattoos	(i) K_gas for lawsone at 298 K	(i) K_gas for lawsone at 298 K	C4 Thermodynamics	P2	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2025 Q1	Clay pigeon shooting: nitrocellulose chemistry	(a)(ii) Dot-and-cross diagram of NO2+	(a)(ii) Dot-and-cross diagram of NO2+	C2 Bonding & Structure	P1	Very High
2025 Q1	Clay pigeon shooting: nitrocellulose chemistry	(b) Formation of the intermediate H2NO3+	(b) Formation of the intermediate H2NO3+	C8 Organic Chemistry	P2	Very High
2025 Q1	Clay pigeon shooting: nitrocellulose chemistry	(c) Decomposition of H2NO3+	(c) Decomposition of H2NO3+	C8 Organic Chemistry	P2	Very High
2025 Q1	Clay pigeon shooting: nitrocellulose chemistry	(e) Standard enthalpy of combustion	(e) Standard enthalpy of combustion	C4 Thermodynamics	P1	Very High
2025 Q3	Epoxides and epoxy resin	(b) C–C–O bond angles (tick table)	(b) C–C–O bond angles (tick table)	C2 Bonding & Structure	P1	Very High
2025 Q3	Epoxides and epoxy resin	(c) Compound A (peroxyacid epoxidation by-product)	(c) Compound A (peroxyacid epoxidation by-product)	C8 Organic Chemistry	P2	Very High
2025 Q3	Epoxides and epoxy resin	(d) Both stereoisomers from propene	(d) Both stereoisomers from propene	C8 Organic Chemistry	P2	Very High
2025 Q3	Epoxides and epoxy resin	(e) Number of epoxide stereoisomers from each C4H8 isomer (tick table)	(e) Number of epoxide stereoisomers from each C4H8 isomer (tick table)	C8 Organic Chemistry	P2	Very High
2025 Q3	Epoxides and epoxy resin	(f) Ring-openings of ethylene oxide: B, C, D	(f) Ring-openings of ethylene oxide: B, C, D	C8 Organic Chemistry	P2	Very High
2025 Q3	Epoxides and epoxy resin	(g) Epichlorohydrin + sodium phenoxide: E- and F	(g) Epichlorohydrin + sodium phenoxide: E- and F	C8 Organic Chemistry	P2	Very High
2025 Q5	Agarwood: the agarospirol synthesis	(a) Stereocentres in agarospirol (circle task)	(a) Stereocentres in agarospirol (circle task)	C8 Organic Chemistry	P2	Very High
2024 Q1	Bronze: configurations, density and the fcc cell	(a) Electronic configuration of a silver atom (tick)	(a) Electronic configuration of a silver atom (tick)	C1 Atomic Structure & Periodicity	P1	Very High
2024 Q1	Bronze: configurations, density and the fcc cell	(b) Electronic configuration of Cu2+ (tick)	(b) Electronic configuration of Cu2+ (tick)	C1 Atomic Structure & Periodicity	P1	Very High
2024 Q2	Iodate salts and the salesite mineral	(b) Dot-and-cross of the iodate ion	(b) Dot-and-cross of the iodate ion	C2 Bonding & Structure	P1	Very High
2024 Q2	Iodate salts and the salesite mineral	(c) O–I–O bond angle (tick)	(c) O–I–O bond angle (tick)	C2 Bonding & Structure	P1	Very High
2024 Q3	Fuel-producing bacteria: ring strain and enolate chemistry	(b)(ii) Enthalpy of formation of cyclopropane	(b)(ii) Enthalpy of formation of cyclopropane	C4 Thermodynamics	P1	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2024 Q3	Fuel-producing bacteria: ring strain and enolate chemistry	(b)(iii) Combustion enthalpy per mole of carbon — cyclopropane	(b)(iii) Combustion enthalpy per mole of carbon — cyclopropane	C4 Thermodynamics	P1	Very High
2024 Q3	Fuel-producing bacteria: ring strain and enolate chemistry	(b)(iv) Per mole of carbon — cyclohexane	(b)(iv) Per mole of carbon — cyclohexane	C4 Thermodynamics	P1	Very High
2024 Q3	Fuel-producing bacteria: ring strain and enolate chemistry	(c) Resonance form of the enolate	(c) Resonance form of the enolate	C8 Organic Chemistry	P2	Very High
2024 Q3	Fuel-producing bacteria: ring strain and enolate chemistry	(d) Enolate + iodomethane (tick statement)	(d) Enolate + iodomethane (tick statement)	C8 Organic Chemistry	P2	Very High
2024 Q3	Fuel-producing bacteria: ring strain and enolate chemistry	(e) Compound A (enolate + carbonyl)	(e) Compound A (enolate + carbonyl)	C8 Organic Chemistry	P2	Very High
2024 Q3	Fuel-producing bacteria: ring strain and enolate chemistry	(f)(ii) Steps involving enolate chemistry (tick)	(f)(ii) Steps involving enolate chemistry (tick)	C8 Organic Chemistry	P2	Very High
2024 Q4	Gadopiclenol: ligand synthesis and relaxation analysis	(b) Stereocentres on H (circle task)	(b) Stereocentres on H (circle task)	C8 Organic Chemistry	P2	Very High
2024 Q5	Sulfur chemistry in the atmosphere	(b)(ii) Structure of D	(b)(ii) Structure of D	C2 Bonding & Structure	P1	Very High
2023 Q1	Rocket fuels	(c)(ii) Energy released on combustion	(c)(ii) Energy released on combustion	C4 Thermodynamics	P1	Very High
2023 Q2	Electronegativity, bonding and BN structures	(a) Pauling Δχ(Cl–H)	(a) Pauling Δχ(Cl–H)	C2 Bonding & Structure	P1	Very High
2023 Q2	Electronegativity, bonding and BN structures	(b) χ(Cl)	(b) χ(Cl)	C1 Atomic Structure & Periodicity	P1	Very High
2023 Q2	Electronegativity, bonding and BN structures	(c) Adjusted Mulliken χ(N)	(c) Adjusted Mulliken χ(N)	C2 Bonding & Structure	P2	Very High
2023 Q2	Electronegativity, bonding and BN structures	(d) Positions on the van Arkel triangle	(d) Positions on the van Arkel triangle	C2 Bonding & Structure	P2	Very High
2023 Q2	Electronegativity, bonding and BN structures	(e) Greatest metallic character	(e) Greatest metallic character	C2 Bonding & Structure	P2	Very High
2023 Q2	Electronegativity, bonding and BN structures	(f) Second plot (Cs, CsBr, CO given as anchors)	(f) Second plot (Cs, CsBr, CO given as anchors)	C2 Bonding & Structure	P2	Very High
2023 Q3	Amino acid complexes	(a) Isoleucine structures	(a) Isoleucine structures	C8 Organic Chemistry	P1	Very High
2023 Q3	Amino acid complexes	(d) Stereoisomers of square-pyramidal Cu(gly)2(H2O)	(d) Stereoisomers of square-pyramidal Cu(gly)2(H2O)	C8 Organic Chemistry	P2	Very High
2023 Q3	Amino acid complexes	(e) Stereoisomers of octahedral MMt3	(e) Stereoisomers of octahedral MMt3	C8 Organic Chemistry	P2	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2023 Q4	Vaping: vitamin E acetate, ketene, and the synthesis	(a)(i) Functional groups (tick)	(a)(i) Functional groups (tick)	C8 Organic Chemistry	P1	Very High
2023 Q4	Vaping: vitamin E acetate, ketene, and the synthesis	(b) The toxic product (yes/no table)	(b) The toxic product (yes/no table)	C8 Organic Chemistry	P2	Very High
2023 Q4	Vaping: vitamin E acetate, ketene, and the synthesis	(c) Wittig reaction table	(c) Wittig reaction table	C8 Organic Chemistry	P2	Very High
2023 Q4	Vaping: vitamin E acetate, ketene, and the synthesis	(e) The intermediate carbocation	(e) The intermediate carbocation	C8 Organic Chemistry	P2	Very High
2023 Q5	Cheese: fermentation equilibria and bubble physics	(b) Internal hydrogen bond in the conjugate base	(b) Internal hydrogen bond in the conjugate base	C2 Bonding & Structure	P1	Very High
2023 Q5	Cheese: fermentation equilibria and bubble physics	(g) [CO2(ch)] at pH 5.20	(g) [CO2(ch)] at pH 5.20	C6 Equilibrium & Acids–Bases	P2	Very High
2022 Q1	E10 petrol	(d) Enthalpy of combustion of octane (bond enthalpies)	(d) Enthalpy of combustion of octane (bond enthalpies)	C4 Thermodynamics	P1	Very High
2022 Q2	Nitrous oxide	(a) Dot-and-cross of HNO2	(a) Dot-and-cross of HNO2	C2 Bonding & Structure	P1	Very High
2022 Q2	Nitrous oxide	(c) Ions B+ and C- with shapes	(c) Ions B+ and C- with shapes	C2 Bonding & Structure	P1	Very High
2022 Q2	Nitrous oxide	(e) Structure of N2O	(e) Structure of N2O	C2 Bonding & Structure	P1	Very High
2022 Q3	Cubane	(d) W and X (Diels–Alder)	(d) W and X (Diels–Alder)	C8 Organic Chemistry	P2	Very High
2022 Q3	Cubane	(e) Starting material Y	(e) Starting material Y	C8 Organic Chemistry	P2	Very High
2022 Q3	Cubane	(f) Intermediate C	(f) Intermediate C	C8 Organic Chemistry	P2	Very High
2022 Q5	The smallest Chinese knot	(a)(i) Structure of A	(a)(i) Structure of A	C8 Organic Chemistry	P2	Very High
2022 Q5	The smallest Chinese knot	(a)(ii) The active electrophile	(a)(ii) The active electrophile	C8 Organic Chemistry	P2	Very High
2022 Q5	The smallest Chinese knot	(b) Structure of B	(b) Structure of B	C8 Organic Chemistry	P2	Very High
2022 Q5	The smallest Chinese knot	(c)(i) Structure of C	(c)(i) Structure of C	C8 Organic Chemistry	P2	Very High
2022 Q5	The smallest Chinese knot	(c)(ii) Structure of D	(c)(ii) Structure of D	C8 Organic Chemistry	P2	Very High
2022 Q5	The smallest Chinese knot	(d) Compound E and reagent G	(d) Compound E and reagent G	C8 Organic Chemistry	P2	Very High
2022 Q5	The smallest Chinese knot	(e) The primary amine RNH2	(e) The primary amine RNH2	C8 Organic Chemistry	P2	Very High
2022 Q6	Storing vaccines: ensilication	(b) ΔH° for SiO2 + 6 NaF + 6 HCl → H2SiF6 + 6 NaCl + 2 H2O	(b) ΔH° for SiO2 + 6 NaF + 6 HCl → H2SiF6 + 6 NaCl + 2 H2O	C4 Thermodynamics	P1	Very High
2022 Q6	Storing vaccines: ensilication	(c) ΔH° and ΔS° for process 2	(c) ΔH° and ΔS° for process 2	C4 Thermodynamics	P2	Very High
2021 Q1	Life on Venus: phosphine	(a) Shape of phosphine	(a) Shape of phosphine	C2 Bonding & Structure	P1	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2021 Q2	Capturing carbon: calcium looping	(f) ΔH° of the air contactor reaction	(f) ΔH° of the air contactor reaction	C4 Thermodynamics	P1	Very High
2021 Q2	Capturing carbon: calcium looping	(h) Entropy changes (tick two)	(h) Entropy changes (tick two)	C4 Thermodynamics	P2	Very High
2021 Q3	Levulinic acid from fructose	(a) Functional groups in levulinic acid (tick)	(a) Functional groups in levulinic acid (tick)	C8 Organic Chemistry	P1	Very High
2021 Q3	Levulinic acid from fructose	(d)(i) Alkene A	(d)(i) Alkene A	C8 Organic Chemistry	P2	Very High
2021 Q3	Levulinic acid from fructose	(e) Intermediates C and D	(e) Intermediates C and D	C8 Organic Chemistry	P2	Very High
2021 Q3	Levulinic acid from fructose	(f) By-product F	(f) By-product F	C8 Organic Chemistry	P2	Very High
2021 Q3	Levulinic acid from fructose	(h) Other stereoisomers of [RuH2(CO)(PPh3)3]	(h) Other stereoisomersof [RuH2(CO)(PPh3)3]	C8 Organic Chemistry	P2	Very High
2021 Q4	'Social distancing' in molecules: conformations	(a) E or S in the boxes (ethane rotation)	(a) E or S in the boxes (ethane rotation)	C8 Organic Chemistry	P2	Very High
2021 Q4	'Social distancing' in molecules: conformations	(b) Methylbutane projections → points A–F	(b) Methylbutane projections → points A–F	C8 Organic Chemistry	P2	Very High
2021 Q4	'Social distancing' in molecules: conformations	(d) Equal-energy sets among 7–12 (acetylcholine)	(d) Equal-energy sets among 7–12 (acetylcholine)	C8 Organic Chemistry	P2	Very High
2021 Q4	'Social distancing' in molecules: conformations	(e) Which of 7–12 are mimicked by cyclopropanes W, X, Y, Z	(e) Which of 7–12 are mimicked by cyclopropanes W, X, Y, Z	C8 Organic Chemistry	P2	Very High
2021 Q5	Hydroxychloroquine synthesis	(b) B and C	(b) B and C	C8 Organic Chemistry	P2	Very High
2021 Q5	Hydroxychloroquine synthesis	(c) Reagents/conditions for step (i) (tick)	(c) Reagents/conditions for step (i) (tick)	C8 Organic Chemistry	P2	Very High
2021 Q5	Hydroxychloroquine synthesis	(e) V-, W+, X+, Y, Z	(e) V-, W+, X+, Y, Z	C8 Organic Chemistry	P2	Very High
2021 Q6	Xenon fluorides	(b) Dot-and-cross of XeF4	(b) Dot-and-cross of XeF4	C2 Bonding & Structure	P1	Very High
2021 Q6	Xenon fluorides	(c) Two arrangements of six pairs in XeF4 (tick)	(c) Two arrangements of six pairs in XeF4 (tick)	C2 Bonding & Structure	P1	Very High
2021 Q6	Xenon fluorides	(d) Arrangements of five pairs in XeF2 (tick)	(d) Arrangements of five pairs in XeF2 (tick)	C2 Bonding & Structure	P1	Very High
2020 Q1	Calcium carbide	(a)(ii) Dot-and-cross of C22-	(a)(ii) Dot-and-cross of C22-	C2 Bonding & Structure	P1	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2020 Q2	Hydrogen as a fuel	(a) Reforming enthalpy	(a) Reforming enthalpy	C4 Thermodynamics	P1	Very High
2020 Q2	Hydrogen as a fuel	(b) Cathode half-reaction	(b) Cathode half-reaction	C7 Electrochemistry	P1	Very High
2020 Q2	Hydrogen as a fuel	(c) Overall cell enthalpy	(c) Overall cell enthalpy	C4 Thermodynamics	P1	Very High
2020 Q2	Hydrogen as a fuel	(d) Cell potential	(d) Cell potential	C7 Electrochemistry	P2	Very High
2020 Q2	Hydrogen as a fuel	(e) Standard electrode potential of half reaction 1	(e) Standard electrode potential of half reaction 1	C7 Electrochemistry	P3	Very High
2020 Q2	Hydrogen as a fuel	(f) ΔcH° of CH4	(f) ΔcH° of CH4	C4 Thermodynamics	P1	Very High
2020 Q2	Hydrogen as a fuel	(g) ΔcH° of H2	(g) ΔcH° of H2	C4 Thermodynamics	P1	Very High
2020 Q3	Sun cream: oxybenzone and octinoxate	(a) Functional groups in oxybenzone (tick)	(a) Functional groups in oxybenzone (tick)	C8 Organic Chemistry	P1	Very High
2020 Q3	Sun cream: oxybenzone and octinoxate	(b) The intramolecular H-bond (circle)	(b) The intramolecular H-bond (circle)	C2 Bonding & Structure	P1	Very High
2020 Q3	Sun cream: oxybenzone and octinoxate	(d) Octinoxate stereoisomers	(d) Octinoxate stereoisomers	C8 Organic Chemistry	P2	Very High
2020 Q3	Sun cream: oxybenzone and octinoxate	(e) E and F	(e) E and F	C8 Organic Chemistry	P2	Very High
2020 Q3	Sun cream: oxybenzone and octinoxate	(f) G and H	(f) G and H	C8 Organic Chemistry	P2	Very High
2020 Q3	Sun cream: oxybenzone and octinoxate	(h) Truxinate stereoisomers (adjacent Ar, adjacent R on the cyclobutane)	(h) Truxinate stereoisomers (adjacent Ar, adjacent R on the cyclobutane)	C8 Organic Chemistry	P2	Very High
2020 Q5	Colourful compounds: conjugation and porphyrins	(a) Skeletal structure of buta-1,3-diene	(a) Skeletal structure of buta-1,3-diene	C8 Organic Chemistry	P1	Very High
2019 Q1	Carbon dioxide	(a)(i) Dot-and-cross diagrams	(a)(i) Dot-and-cross diagrams	C2 Bonding & Structure	P1	Very High
2019 Q1	Carbon dioxide	(c) Maximum safe storage temperature	(c) Maximum safe storage temperature	C6 Equilibrium & Acids–Bases	P2	Very High
2019 Q1	Carbon dioxide	(e) Enthalpy of the WGS reaction	(e) Enthalpy of the WGS reaction	C4 Thermodynamics	P1	Very High
2019 Q2	Industrial separation of precious metals	(a) Electronic structures	(a) Electronic structures	C1 Atomic Structure & Periodicity	P1	Very High
2019 Q3	Treating nerve agent poisoning	(e) (E)-1,4-dibromobut-2-ene	(e) (E)-1,4-dibromobut-2-ene	C8 Organic Chemistry	P1	Very High
2019 Q3	Treating nerve agent poisoning	(g) X and Y	(g) X and Y	C8 Organic Chemistry	P2	Very High
2019 Q4	Bees and Brexit: thiamethoxam	(c) Thiocyanate resonance explaining C	(c) Thiocyanate resonance explaining C	C8 Organic Chemistry	P2	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2019 Q4	Bees and Brexit: thiamethoxam	(d) The electrophile for guanidine → E	(d) The electrophile for guanidine → E	C8 Organic Chemistry	P2	Very High
2019 Q4	Bees and Brexit: thiamethoxam	(g) Lowest-energy tautomer of K	(g) Lowest-energy tautomer of K	C8 Organic Chemistry	P2	Very High
2019 Q4	Bees and Brexit: thiamethoxam	(h) L and M	(h) L and M	C8 Organic Chemistry	P2	Very High
2019 Q5	A biodegradable plastic: PLA	(a) Lactic acid and its chiral centre	(a) Lactic acid and its chiral centre	C8 Organic Chemistry	P1	Very High
2018 Q1	Lithium compounds	(c) Structure of EC (C3H4O3)	(c) Structure of EC (C3H4O3)	C8 Organic Chemistry	P1	Very High
2018 Q2	Making ammonia by lithium cycling	(a) Half-equations of Step 1	(a) Half-equations of Step 1	C7 Electrochemistry	P1	Very High
2018 Q2	Making ammonia by lithium cycling	(b) ΔH°, ΔS°, ΔG° for Step 1 at 750 K	(b) ΔH°, ΔS°, ΔG° for Step 1 at 750 K	C4 Thermodynamics	P2	Very High
2018 Q3	Enriched uranium	(d) Is UF6 polar? (circle)	(d) Is UF6 polar? (circle)	C2 Bonding & Structure	P1	Very High
2018 Q4	Cough suppressants: dextromethorphan	(d) Anions of cyanoacetic acid	(d) Anions of cyanoacetic acid	C8 Organic Chemistry	P2	Very High
2018 Q4	Cough suppressants: dextromethorphan	(e) The two carbons connected in F → G	(e) The two carbons connected in F → G	C8 Organic Chemistry	P2	Very High
2018 Q4	Cough suppressants: dextromethorphan	(f) H and cation I+	(f) H and cation I+	C8 Organic Chemistry	P2	Very High
2018 Q4	Cough suppressants: dextromethorphan	(g) Pairs of carbons that can be connected in J → M1/M2	(g) Pairs of carbons that can be connected in J → M1/M2	C8 Organic Chemistry	P2	Very High
2018 Q4	Cough suppressants: dextromethorphan	(i) M1 and M2	(i) M1 and M2	C8 Organic Chemistry	P2	Very High
2018 Q5	The 'inert' gas helium	(h) Minimum pressure for favourable formation	(h) Minimum pressure for favourable formation	C4 Thermodynamics	P2	Very High
2017 Q1	The Green Pool of Rio	(e)(ii) Structure of NCl3	(e)(ii) Structure of NCl3	C2 Bonding & Structure	P1	Very High
2016 Q1	Energy storage: the sulfur–iodine cycle	(c) ΔrH° for reaction (1)	(c) ΔrH° for reaction (1)	C4 Thermodynamics	P1	Very High
2016 Q4	The synthesis of Addyi (flibanserin)	(e) Intermediate X and compound Y	(e) Intermediate X and compound Y	C8 Organic Chemistry	P2	Very High
2015 Q2	Detecting molecules in space	(a) The branched nitrile and an isomer	(a) The branched nitrile and an isomer	C8 Organic Chemistry	P1	Very High
2015 Q3	Ritalin®	(a) Compound A	(a) Compound A	C8 Organic Chemistry	P2	Very High
2015 Q3	Ritalin®	(b) Anion B-	(b) Anion B-	C8 Organic Chemistry	P2	Very High
2015 Q3	Ritalin®	(h) Anion G-	(h) Anion G-	C8 Organic Chemistry	P2	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2015 Q3	Ritalin®	(i) Isomers interconvertible via G-	(i) Isomers interconvertible via G-	C8 Organic Chemistry	P2	Very High
2015 Q5	"Green" jet fuel	(h) Combustion enthalpy of dodecane and the FT reaction enthalpy	(h) Combustion enthalpy of dodecane and the FT reaction enthalpy	C4 Thermodynamics	P1	Very High
2014 Q1	Controlling phosphate levels	(a) Lanthanum carbonate with stomach acid; phosphate precipitation	(a) Lanthanum carbonate with stomach acid; phosphate precipitation	C6 Equilibrium & Acids–Bases	P1	Very High
2014 Q2	A sodium street lamp	(a)–(b) Electronic configuration and valence electron	(a)–(b) Electronic configuration and valence electron	C1 Atomic Structure & Periodicity	P1	Very High
2014 Q3	Spot cream: the synthesis of tazarotene	(b) Compound A and Ester B	(b) Compound A and Ester B	C8 Organic Chemistry	P2	Very High
2014 Q3	Spot cream: the synthesis of tazarotene	(e) Anion J	(e) Anion J	C8 Organic Chemistry	P2	Very High
2014 Q4	Bombardier beetles	(d) Counting 1,2,4-trisubstitution isomers	(d) Counting 1,2,4-trisubstitution isomers	C8 Organic Chemistry	P2	Very High
2014 Q5	Fire and ice: methane clathrates	(f) Enthalpy of formation of the hydrate	(f) Enthalpy of formation of the hydrate	C4 Thermodynamics	P1	Very High
2013 Q1	Rocket fuels	(b) Enthalpy of decomposition of hydrazine to its elements	(b) Enthalpy of decomposition of hydrazine to its elements	C4 Thermodynamics	P1	Very High
2013 Q3	Coloured compounds	(a) Colour assignments	(a) Colour assignments	C12 Clearly Beyond the Syllabus	P4	Very High
2012 Q1	Sulfur chemistry	(a) Enthalpy of 8 S7(g) → 7 S8(g)	(a) Enthalpy of 8 S7(g) → 7 S8(g)	C4 Thermodynamics	P1	Very High
2012 Q1	Sulfur chemistry	(b) Solution equilibrium of S7 and S8	(b) Solution equilibrium of S7 and S8	C6 Equilibrium & Acids–Bases	P2	Very High
2012 Q1	Sulfur chemistry	(d) S4N4 without cross-links	(d) S4N4 without cross-links	C2 Bonding & Structure	P2	Very High
2012 Q1	Sulfur chemistry	(e) The cross-linked alternative	(e) The cross-linked alternative	C2 Bonding & Structure	P2	Very High
2012 Q1	Sulfur chemistry	(f) Poly(sulfur nitride), (SN)x	(f) Poly(sulfur nitride), (SN)x	C2 Bonding & Structure	P2	Very High
2012 Q2	Controlling your hair	(c) Predominant forms of thioglycolic acid	(c) Predominant forms of thioglycolic acid	C6 Equilibrium & Acids–Bases	P1	Very High
2012 Q3	Chemistry general knowledge	(b) Boiling-point ranking	(b) Boiling-point ranking	C2 Bonding & Structure	P1	Very High
2012 Q4	The synthesis of Lipitor® (atorvastatin)	(b) Anion I	(b) Anion I	C8 Organic Chemistry	P2	Very High
2012 Q4	The synthesis of Lipitor® (atorvastatin)	(d) Phenylamine and 4-fluorobenzaldehyde	(d) Phenylamine and 4-fluorobenzaldehyde	C8 Organic Chemistry	P2	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2012 Q4	The synthesis of Lipitor® (atorvastatin)	(f) Compound U	(f) Compound U	C8 Organic Chemistry	P2	Very High
2012 Q5	The world's smallest powered car	(b) R/S assignment of the rotor units (order: top-left, bottom-left, top-right, bottom-right)	(b) R/S assignment of the rotor units (order: top-left, bottom-left, top-right, bottom-right)	C8 Organic Chemistry	P2	Very High
2012 Q5	The world's smallest powered car	(c) Which cars are optical isomers	(c) Which cars are optical isomers	C8 Organic Chemistry	P2	Very High
2012 Q5	The world's smallest powered car	(d) Cars that are the same molecule when free from the surface	(d) Cars that are the same molecule when free from the surface	C8 Organic Chemistry	P2	Very High
2011 Q1	A popular snack food	(e) Chiral centres in citric acid	(e) Chiral centres in citric acid	C8 Organic Chemistry	P1	Very High
2011 Q2	Calcium carbide	(a) Dot-cross diagram of C22-	(a) Dot-cross diagram of C22-	C2 Bonding & Structure	P1	Very High
2011 Q2	Calcium carbide	(b) Diatomic gases isoelectronic with C22-	(b) Diatomic gases isoelectronic with C22-	C2 Bonding & Structure	P1	Very High
2010 Q4	Salting roads in winter	(d) Freezing point of 3.00 M NaCl and the solubility limit	(d) Freezing point of 3.00 M NaCl and the solubility limit	C6 Equilibrium & Acids–Bases	P2	Very High
2010 Q5	Osmium in organic reactions	(e) The four stereoisomers of pentane-2,3-diol — required alkenes	(e) The four stereoisomers of pentane-2,3-diol — required alkenes	C8 Organic Chemistry	P2	Very High
2010 Q5	Osmium in organic reactions	(g) The sixteen cyclic-ether stereoisomers	(g) The sixteen cyclic-ether stereoisomers	C8 Organic Chemistry	P2	Very High
2009 Q4	Loperamide (anti-diarrhoea)	(b) Ester A and anion B	(b) Ester A and anion B	C8 Organic Chemistry	P2	Very High
2009 Q4	Loperamide (anti-diarrhoea)	(c) Anion C and cyclic ester D	(c) Anion C and cyclic ester D	C8 Organic Chemistry	P2	Very High
2008 Q2	Vitamin D production in mushrooms	(a) Chiral centres of ergosterol	(a) Chiral centres of ergosterol	C8 Organic Chemistry	P1	Very High
2008 Q3	Mercury fulminate	(b) Enthalpy of detonation	(b) Enthalpy of detonation	C4 Thermodynamics	P1	Very High
2008 Q3	Mercury fulminate	(c) Structure of mercury(II) fulminate	(c) Structure of mercury(II) fulminate	C2 Bonding & Structure	P2	Very High
2008 Q3	Mercury fulminate	(d) Structure of mercury(II) cyanate	(d) Structure of mercury(II) cyanate	C2 Bonding & Structure	P2	Very High
2008 Q3	Mercury fulminate	(e) The two forms of cyanuric acid	(e) The two forms of cyanuric acid	C8 Organic Chemistry	P2	Very High
2008 Q4	Aluminium chemistry and rat poison	(a) Electrons around Al in covalent AlCl3	(a) Electrons around Al in covalent AlCl3	C2 Bonding & Structure	P1	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2008 Q4	Aluminium chemistry and rat poison	(b) Vapour-phase molecule A (M_r = 266.66)	(b) Vapour-phase molecule A (M_r = 266.66)	C2 Bonding & Structure	P2	Very High
2008 Q4	Aluminium chemistry and rat poison	(d) Electrons around Al in iBu2AlH	(d) Electrons around Al in iBu2AlH	C2 Bonding & Structure	P1	Very High
2008 Q6	The synthesis of fexofenadine	(b) Starting material X and the base's targets	(b) Starting material X and the base's targets	C8 Organic Chemistry	P2	Very High
2007 Q1	Launching the space shuttle	(d) Enthalpy of the booster reaction	(d) Enthalpy of the booster reaction	C4 Thermodynamics	P1	Very High
2007 Q2	Geometric shapes in chemistry	(a)–(b) White phosphorus P4	(a)–(b) White phosphorus P4	C2 Bonding & Structure	P1	Very High
2007 Q2	Geometric shapes in chemistry	(c) The two oxides	(c) The two oxides	C2 Bonding & Structure	P1	Very High
2007 Q2	Geometric shapes in chemistry	(d) Phosphorus(III) oxide	(d) Phosphorus(III) oxide	C2 Bonding & Structure	P1	Very High
2007 Q2	Geometric shapes in chemistry	(e) Phosphorus(V) oxide	(e) Phosphorus(V) oxide	C2 Bonding & Structure	P1	Very High
2007 Q2	Geometric shapes in chemistry	(f)–(g) Phosphoric(V) acid	(f)–(g) Phosphoric(V) acid	C2 Bonding & Structure	P1	Very High
2007 Q3	Phosphate levels in our environment	(c) Prevalent phosphate form at pH 7.4	(c) Prevalent phosphate form at pH 7.4	C6 Equilibrium & Acids–Bases	P1	Very High
2007 Q3	Phosphate levels in our environment	(f) Which phosphate precipitates first	(f) Which phosphate precipitates first	C6 Equilibrium & Acids–Bases	P2	Very High
2007 Q4	The flame-retardant cation (THP+)	(a) Phosphine and methanal geometries	(a) Phosphine and methanal geometries	C2 Bonding & Structure	P1	Very High
2007 Q5	The synthesis of Tamiflu (Rohloff route)	(b) Chiral centres and stereoisomer count	(b) Chiral centres and stereoisomer count	C8 Organic Chemistry	P2	Very High
2007 Q6	Chlorine dioxide	(a) ClO2 dot-cross diagram and angle	(a) ClO2 dot-cross diagram and angle	C2 Bonding & Structure	P1	Very High
2006 Q1	Sherbet lemons	(e) Stereoisomers of tartaric acid	(e) Stereoisomers of tartaric acid	C8 Organic Chemistry	P2	Very High
2006 Q1	Sherbet lemons	(f) Asymmetric carbons in citric acid	(f) Asymmetric carbons in citric acid	C8 Organic Chemistry	P1	Very High
2006 Q2	Redox chemistry of vehicle pollution	(f) Urea structure and bond angles	(f) Urea structure and bond angles	C2 Bonding & Structure	P1	Very High
2006 Q3	Acyl chlorides and related oxochlorides	(a)–(b) Acyl chloride reactions	(a)–(b) Acyl chloride reactions	C8 Organic Chemistry	P1	Very High
2006 Q6	The synthesis of Viagra (sildenafil)	(a) Intermediates B, D, E, F, G, I, J	(a) Intermediates B, D, E, F, G, I, J	C8 Organic Chemistry	P2	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2006 Q6	The synthesis of Viagra (sildenafil)	(b) Reagents for B → C and E → F	(b) Reagents for B → C and E → F	C8 Organic Chemistry	P2	Very High
2005 Q1	Carbon oxides	(a)(ii) Structure of carbonic acid	(a)(ii) Structure of carbonic acid	C2 Bonding & Structure	P1	Very High
2005 Q1	Carbon oxides	(b)(ii) Dot-and-cross structure of CO	(b)(ii) Dot-and-cross structure of CO	C2 Bonding & Structure	P1	Very High
2005 Q1	Carbon oxides	(b)(iii) Bond order in CO	(b)(iii) Bond order in CO	C2 Bonding & Structure	P2	Very High
2005 Q1	Carbon oxides	(d)(ii) Structure of carbon suboxide	(d)(ii) Structure of carbon suboxide	C2 Bonding & Structure	P1	Very High
2005 Q2	Diiodine pentoxide	(e) Structures of HIO3 and I2O5	(e) Structures of HIO3 and I2O5	C2 Bonding & Structure	P1	Very High
2005 Q5	The Breathalyser	(a)(i) Oxidation of ethanol	(a)(i) Oxidation of ethanol	C7 Electrochemistry	P1	Very High
2005 Q5	The Breathalyser	(a)(ii) Reduction of dichromate	(a)(ii) Reduction of dichromate	C7 Electrochemistry	P1	Very High
2004 Q1	Redox half-equations (in aqueous acid)	(whole question)	balancing redox half-equations in acidic solution	C7 Electrochemistry	P1	Very High
2004 Q2	Pollution and the Taj Mahal	(b)(ii) Why the conversion accelerates erosion	(b)(ii) Why the conversion accelerates erosion	C6 Equilibrium & Acids–Bases	P2	Very High
2004 Q2	Pollution and the Taj Mahal	(c) Hydrolysis of urea	(c) Hydrolysis of urea	C8 Organic Chemistry	P2	Very High
2004 Q2	Pollution and the Taj Mahal	(d)(ii) Why BaCO3 slows erosion	(d)(ii) Why BaCO3 slows erosion	C6 Equilibrium & Acids–Bases	P2	Very High
2004 Q4	Agent Orange	(a) 1,2,4,5-tetrachlorobenzene	(a) 1,2,4,5-tetrachlorobenzene	C8 Organic Chemistry	P1	Very High
2004 Q4	Agent Orange	(b) Intermediate X	(b) Intermediate X	C8 Organic Chemistry	P2	Very High
2004 Q4	Agent Orange	(e) Chloroethanoic acid	(e) Chloroethanoic acid	C8 Organic Chemistry	P1	Very High
2004 Q5	Tin pest	(a) ΔrH for white → grey	(a) ΔrH for white → grey	C4 Thermodynamics	P1	Very High
2004 Q5	Tin pest	(b) ΔrS	(b) ΔrS	C4 Thermodynamics	P2	Very High
2004 Q5	Tin pest	(c) ΔrG at 25 °C	(c) ΔrG at 25 °C	C4 Thermodynamics	P2	Very High
2004 Q5	Tin pest	(e) Critical temperature Tc	(e) Critical temperature Tc	C4 Thermodynamics	P2	Very High
2004 Q7	Asparagus-pee	(b) Structures (all SMILES machine-validated)	(b) Structures (all SMILES machine-validated)	C8 Organic Chemistry	P2	Very High
2003 Q1	Heating a cup of coffee	(c) ΔrH of the reaction	(c) ΔrH of the reaction	C4 Thermodynamics	P1	Very High
2003 Q2	Reinecke's salt, NH4[Cr(SCN)x(NH3)γ]	(c) Shape of the anion	(c) Shape of the anion	C2 Bonding & Structure	P1	Very High
2003 Q2	Reinecke's salt, NH4[Cr(SCN)x(NH3)γ]	(d) Two structures + isomerism	(d) Two structures + isomerism	C8 Organic Chemistry	P2	Very High

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2003 Q4	Redox equations (half-equation method)	(whole question)	balancing redox equations by the half-reaction method	C7 Electrochemistry	P1	Very High
2003 Q6	Hydroxylamine and iron(III)	(a) Structure of NH3OH+ Cl-	(a) Structure of NH3OH+ Cl-	C2 Bonding & Structure	P1	Very High
2003 Q8	Polonium	(a) Electronic configuration	(a) Electronic configuration	C1 Atomic Structure & Periodicity	P1	Very High

II-C.2 Low & Very Low Frequency (203 items) — safe to deprioritise

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2026 Q2	Isocyanides	(e) Zaytsev (carbylamine) equation	(e) Zaytsev (carbylamine) equation	C8 Organic Chemistry	P2	Low
2026 Q2	Isocyanides	(f) Which compounds give a positive Zaytsev test	(f) Which compounds give a positive Zaytsev test	C8 Organic Chemistry	P2	Low
2026 Q4	Rice, spice, and mice	(c) Ordering the butadiene MOs A–D	(c) Ordering the butadiene MOs A–D	C12 Clearly Beyond the Syllabus	P4	Very Low
2026 Q4	Rice, spice, and mice	(d) Completing the hexatriene MO diagram	(d) Completing the hexatriene MO diagram	C12 Clearly Beyond the Syllabus	P4	Very Low
2026 Q4	Rice, spice, and mice	(e)(i) Energy gap of hexatriene; (ii) absorbed wavelength	(e)(i) Energy gap of hexatriene; (ii) absorbed wavelength	C12 Clearly Beyond the Syllabus	P4	Very Low
2026 Q4	Rice, spice, and mice	(f) Does hexatriene absorb visible light?	(f) Does hexatriene absorb visible light?	C12 Clearly Beyond the Syllabus	P4	Very Low
2026 Q4	Rice, spice, and mice	(g) Trend of λ with conjugation length	(g) Trend of λ with conjugation length	C12 Clearly Beyond the Syllabus	P4	Very Low
2026 Q5	Minerals	(c) Identifying pure samples	(c) Identifying pure samples	C11 Experiments & Graphs	P3	Low
2026 Q5	Minerals	(d) Volume of the PO4 tetrahedron	(d) Volume of the PO4 tetrahedron	C3 Stoichiometry	P1	Low
2026 Q5	Minerals	(e)(i) Unit-cell volume of variscite; (ii) density	(e)(i) Unit-cell volume of variscite; (ii) density	C12 Clearly Beyond the Syllabus	P4	Very Low
2026 Q5	Minerals	(f)(i) Unit-cell volume of turquoise; (ii) density	(f)(i) Unit-cell volume of turquoise; (ii) density	C12 Clearly Beyond the Syllabus	P4	Very Low
2025 Q2	BrAt: astatine chemistry	(a) Natural abundance of At atoms	(a) Natural abundance of At atoms	C10 Inorganic Chemistry	P3	Low

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2025 Q2	BrAt: astatine chemistry	(c) Chlorination products B and C	(c) Chlorination products B and C	C10 Inorganic Chemistry	P3	Low
2025 Q2	BrAt: astatine chemistry	(d) The astatine isotope	(d) The astatine isotope	C3 Stoichiometry	P1	Low
2025 Q2	BrAt: astatine chemistry	(e) Bi + concentrated HNO3	(e) Bi + concentrated HNO3	C10 Inorganic Chemistry	P3	Low
2025 Q2	BrAt: astatine chemistry	(f) D and E	(f) D and E	C10 Inorganic Chemistry	P3	Low
2025 Q2	BrAt: astatine chemistry	(g) Length of BrAt Bear Summer	(g) Length of BrAt Bear Summer	C3 Stoichiometry	P1	Low
2025 Q4	Carbon nanotubes and Raman spectroscopy	(g) The nanotube calibration	(g) The nanotube calibration	C11 Experiments & Graphs	P3	Low
2024 Q1	Bronze: configurations, density and the fcc cell	(e) Side of the copper unit cell	(e) Side of the copper unit cell	C12 Clearly Beyond the Syllabus	P4	Very Low
2024 Q1	Bronze: configurations, density and the fcc cell	(f) Volume of the unit cell	(f) Volume of the unit cell	C12 Clearly Beyond the Syllabus	P4	Very Low
2024 Q1	Bronze: configurations, density and the fcc cell	(g) Density of copper	(g) Density of copper	C12 Clearly Beyond the Syllabus	P4	Very Low
2024 Q2	Iodate salts and the salesite mineral	(j) Anions excluded by "no effervescence"	(j) Anions excluded by "no effervescence"	C10 Inorganic Chemistry	P3	Low
2024 Q2	Iodate salts and the salesite mineral	(k) Identify Mn+ and Zm-	(k) Identify Mn+ and Zm-	C10 Inorganic Chemistry	P3	Low
2024 Q4	Gadopiclenol: ligand synthesis and relaxation analysis	(e) Type of electromagnetic radiation (tick)	(e) Type of electromagnetic radiation (tick)	C11 Experiments & Graphs	P3	Low
2024 Q4	Gadopiclenol: ligand synthesis and relaxation analysis	(f) Net magnetisation at t = 3τ	(f) Net magnetisation at t = 3τ	C12 Clearly Beyond the Syllabus	P4	Very Low
2024 Q4	Gadopiclenol: ligand synthesis and relaxation analysis	(h) Field-independent relaxation time	(h) Field-independent relaxation time	C12 Clearly Beyond the Syllabus	P4	Very Low
2024 Q4	Gadopiclenol: ligand synthesis and relaxation analysis	(j) Value of Aqχ	(j) Value of Aqχ	C12 Clearly Beyond the Syllabus	P4	Very Low
2024 Q4	Gadopiclenol: ligand synthesis and relaxation analysis	(k) Number of coordinated waters q	(k) Number of coordinated waters q	C12 Clearly Beyond the Syllabus	P4	Very Low
2024 Q5	Sulfur chemistry in the atmosphere	(c) Formula of G	(c) Formula of G	C10 Inorganic Chemistry	P3	Low
2024 Q5	Sulfur chemistry in the atmosphere	(f) Formula of L•	(f) Formula of L•	C10 Inorganic Chemistry	P3	Low
2023 Q1	Rocket fuels	(d)(i) Sabatier process	(d)(i) Sabatier process	C10 Inorganic Chemistry	P3	Low
2023 Q2	Electronegativity, bonding and BN structures	(g)(i) Reaction (1): boric acid + ammonia	(g)(i) Reaction (1): boric acid + ammonia	C10 Inorganic Chemistry	P3	Low
2023 Q2	Electronegativity, bonding and BN structures	(g)(ii) Reaction (2): B2O3 + N2 + CaB6	(g)(ii) Reaction (2): B2O3 + N2 + CaB6	C10 Inorganic Chemistry	P3	Low

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2023 Q2	Electronegativity, bonding and BN structures	(h) BN unit cells	(h) BN unit cells	C12 Clearly Beyond the Syllabus	P4	Very Low
2023 Q3	Amino acid complexes	(f) RhMt3 and CuMt2	(f) RhMt3 and CuMt2	C10 Inorganic Chemistry	P3	Low
2023 Q3	Amino acid complexes	(g) FeMt3 spin state	(g) FeMt3 spin state	C12 Clearly Beyond the Syllabus	P4	Very Low
2023 Q5	Cheese: fermentation equilibria and bubble physics	(i) The (a + br3)(p_atm + γ/r) = η form	(i) The (a + br3)(p_atm + γ/r) = η form	C12 Clearly Beyond the Syllabus	P4	Very Low
2023 Q5	Cheese: fermentation equilibria and bubble physics	(j) Two roots and the physical choice	(j) Two roots and the physical choice	C12 Clearly Beyond the Syllabus	P4	Very Low
2022 Q2	Nitrous oxide	(b) Formula of A	(b) Formula of A	C10 Inorganic Chemistry	P3	Low
2022 Q2	Nitrous oxide	(d) D, E, F, X	(d) D, E, F, X	C10 Inorganic Chemistry	P3	Low
2022 Q4	Coronavirus lateral flow test	(e)(i) Antibodies on the test strip	(e)(i) Antibodies on the test strip	C11 Experiments & Graphs	P3	Low
2022 Q4	Coronavirus lateral flow test	(f) Nanoparticles immobilised on the test strip	(f) Nanoparticles immobilised on the test strip	C11 Experiments & Graphs	P3	Low
2022 Q5	The smallest Chinese knot	(f) Metal ions and coordinating atoms on the strand	(f) Metal ions and coordinating atoms on the strand	C10 Inorganic Chemistry	P3	Low
2022 Q5	The smallest Chinese knot	(g) Coordination geometry (tick)	(g) Coordination geometry (tick)	C12 Clearly Beyond the Syllabus	P4	Very Low
2022 Q5	The smallest Chinese knot	(h) M2+ and X-	(h) M2+ and X-	C10 Inorganic Chemistry	P3	Low
2022 Q5	The smallest Chinese knot	(i) Which structures form from each connection pattern	(i) Which structures form from each connection pattern	C12 Clearly Beyond the Syllabus	P4	Very Low
2022 Q6	Storing vaccines: ensilication	(a) Processes 1 and 2	(a) Processes 1 and 2	C10 Inorganic Chemistry	P3	Low
2022 Q6	Storing vaccines: ensilication	(f) Radii r1 and r2	(f) Radii r1 and r2	C12 Clearly Beyond the Syllabus	P4	Very Low
2021 Q1	Life on Venus: phosphine	(c) Three phosphine preparations	(c) Three phosphine preparations	C10 Inorganic Chemistry	P3	Low
2021 Q1	Life on Venus: phosphine	(d) The phosphonium salt	(d) The phosphonium salt	C10 Inorganic Chemistry	P3	Low
2021 Q2	Capturing carbon: calcium looping	(a) Carbonator	(a) Carbonator	C10 Inorganic Chemistry	P3	Low
2021 Q2	Capturing carbon: calcium looping	(b) Calciner	(b) Calciner	C10 Inorganic Chemistry	P3	Low
2021 Q2	Capturing carbon: calcium looping	(c) H2S combustion	(c) H2S combustion	C10 Inorganic Chemistry	P3	Low
2021 Q2	Capturing carbon: calcium looping	(d) SO2 + CaCO3 + O2	(d) SO2 + CaCO3 + O2	C10 Inorganic Chemistry	P3	Low

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2021 Q2	Capturing carbon: calcium looping	(e) Pellet reactor	(e) Pellet reactor	C10 Inorganic Chemistry	P3	Low
2021 Q3	Levulinic acid from fructose	(g) Z in [RuH2(CO)(PPh3)3]	(g) Z in [RuH2(CO)(PPh3)3]	C12 Clearly Beyond the Syllabus	P4	Very Low
2021 Q6	Xenon fluorides	(a) Formation of XeF4	(a) Formation of XeF4	C10 Inorganic Chemistry	P3	Low
2020 Q1	Calcium carbide	(a)(i) Formation of calcium carbide	(a)(i) Formation of calcium carbide	C10 Inorganic Chemistry	P3	Low
2020 Q1	Calcium carbide	(b) Ethyne from calcium carbide	(b) Ethyne from calcium carbide	C10 Inorganic Chemistry	P3	Low
2020 Q1	Calcium carbide	(d) Atoms in the CaC2 unit cell	(d) Atoms in the CaC2 unit cell	C12 Clearly Beyond the Syllabus	P4	Very Low
2020 Q1	Calcium carbide	(e) Cell parameter z	(e) Cell parameter z	C12 Clearly Beyond the Syllabus	P4	Very Low
2020 Q4	Silicon oxides and silicates	(a)(i) Gas Y	(a)(i) Gas Y	C10 Inorganic Chemistry	P3	Low
2020 Q4	Silicon oxides and silicates	(a)(ii) Anion of X	(a)(ii) Anion of X	C10 Inorganic Chemistry	P3	Low
2020 Q4	Silicon oxides and silicates	(a)(iii) Equation	(a)(iii) Equation	C10 Inorganic Chemistry	P3	Low
2020 Q4	Silicon oxides and silicates	(b) CaSiO3 + dilute HCl	(b) CaSiO3 + dilute HCl	C10 Inorganic Chemistry	P3	Low
2020 Q4	Silicon oxides and silicates	(c) Quartz connectivity	(c) Quartz connectivity	C10 Inorganic Chemistry	P3	Low
2020 Q4	Silicon oxides and silicates	(d) Silicate anion structures	(d) Silicate anion structures	C10 Inorganic Chemistry	P3	Low
2020 Q4	Silicon oxides and silicates	(e) Silicate anion in talc	(e) Silicate anion in talc	C10 Inorganic Chemistry	P3	Low
2020 Q5	Colourful compounds: conjugation and porphyrins	(b)(i) ΔE for the UV absorption	(b)(i) ΔE for the UV absorption	C12 Clearly Beyond the Syllabus	P4	Very Low
2020 Q5	Colourful compounds: conjugation and porphyrins	(b)(ii) Length L	(b)(ii) Length L	C12 Clearly Beyond the Syllabus	P4	Very Low
2020 Q5	Colourful compounds: conjugation and porphyrins	(c) Porphine/porphyrin scaling λ = εd2	(c) Porphine/porphyrin scaling λ = εd2	C12 Clearly Beyond the Syllabus	P4	Very Low
2020 Q6	Anammox and ladderanes	(b) The three steps	(b) The three steps	C10 Inorganic Chemistry	P3	Low
2019 Q2	Industrial separation of precious metals	(b) Aqua regia dissolutions	(b) Aqua regia dissolutions	C10 Inorganic Chemistry	P3	Low
2019 Q2	Industrial separation of precious metals	(c) AuCl4- molecular-ion cluster	(c) AuCl4- molecular-ion cluster	C12 Clearly Beyond the Syllabus	P4	Very Low
2019 Q2	Industrial separation of precious metals	(e) Formula of compound X	(e) Formula of compound X	C10 Inorganic Chemistry	P3	Low
2019 Q2	Industrial separation of precious metals	(f) Structures of compound Y	(f) Structures of compound Y	C10 Inorganic Chemistry	P3	Low
2019 Q2	Industrial separation of precious metals	(g) Formation of Y from H2PdCl4	(g) Formation of Y from H2PdCl4	C10 Inorganic Chemistry	P3	Low

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2019 Q3	Treating nerve agent poisoning	(f) Number of different reactivators	(f) Number of different reactivators	C8 Organic Chemistry	P2	Low
2018 Q1	Lithium compounds	(e) $Li_2O_2 + CO_2$ (air purification)	(e) $Li_2O_2 + CO_2$ (air purification)	C10 Inorganic Chemistry	P3	Low
2018 Q1	Lithium compounds	(f) Compound A (oxygen storage)	(f) Compound A (oxygen storage)	C10 Inorganic Chemistry	P3	Low
2018 Q3	Enriched uranium	(a) Energy released per mole of 235U	(a) Energy released per mole of 235U	C12 Clearly Beyond the Syllabus	P4	Very Low
2018 Q3	Enriched uranium	(b) Isotope abundances in the crust	(b) Isotope abundances in the crust	C3 Stoichiometry	P1	Low
2018 Q3	Enriched uranium	(c) Essential fluorine property for centrifuge separation (circle)	(c) Essential fluorine property for centrifuge separation (circle)	C10 Inorganic Chemistry	P3	Low
2018 Q3	Enriched uranium	(f) Back-conversion to uranium metal	(f) Back-conversion to uranium metal	C10 Inorganic Chemistry	P3	Low
2018 Q3	Enriched uranium	(h) R, T, X, Z in the $U_3O_8 \rightarrow UF_6$ scheme	(h) R, T, X, Z in the $U_3O_8 \rightarrow UF_6$ scheme	C10 Inorganic Chemistry	P3	Low
2018 Q3	Enriched uranium	(i) Alpha-decay equations	(i) Alpha-decay equations	C12 Clearly Beyond the Syllabus	P4	Low
2018 Q3	Enriched uranium	(j) Half-lives from the graph	(j) Half-lives from the graph	C5 Kinetics	P2	Low
2018 Q3	Enriched uranium	(k) Age of the Earth	(k) Age of the Earth	C3 Stoichiometry	P1	Low
2018 Q5	The 'inert' gas helium	(c) Atoms in the unit cell of X	(c) Atoms in the unit cell of X	C12 Clearly Beyond the Syllabus	P4	Very Low
2018 Q5	The 'inert' gas helium	(d) Formula of compound X	(d) Formula of compound X	C12 Clearly Beyond the Syllabus	P4	Very Low
2018 Q5	The 'inert' gas helium	(e) Density of X	(e) Density of X	C12 Clearly Beyond the Syllabus	P4	Very Low
2018 Q5	The 'inert' gas helium	(f) What is X? (circle all)	(f) What is X? (circle all)	C12 Clearly Beyond the Syllabus	P4	Very Low
2018 Q5	The 'inert' gas helium	(g) Decomposition of X	(g) Decomposition of X	C10 Inorganic Chemistry	P3	Low
2017 Q1	The Green Pool of Rio	(e)(i) Formation of nitrogen trichloride	(e)(i) Formation of nitrogen trichloride	C10 Inorganic Chemistry	P3	Low
2017 Q1	The Green Pool of Rio	(f) Hydrazine from ammonia + hypochlorite	(f) Hydrazine from ammonia + hypochlorite	C10 Inorganic Chemistry	P3	Low
2017 Q1	The Green Pool of Rio	(g) The blue precipitate on Ryan Lochte's hair	(g) The blue precipitate on Ryan Lochte's hair	C10 Inorganic Chemistry	P3	Low

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2017 Q2	Atmospheric chemistry	(b) Laboratory generation of H2S	(b) Laboratory generation of H2S	C10 Inorganic Chemistry	P3	Low
2017 Q2	Atmospheric chemistry	(c) Atmospheric [H2S]	(c) Atmospheric [H2S]	C3 Stoichiometry	P1	Low
2017 Q2	Atmospheric chemistry	(d) Convert to µg m-3	(d) Convert to µg m-3	C3 Stoichiometry	P1	Low
2017 Q2	Atmospheric chemistry	(e) Partial combustion and Claus chemistry	(e) Partial combustion and Claus chemistry	C10 Inorganic Chemistry	P3	Low
2017 Q2	Atmospheric chemistry	(f) Half-life of 1,1,1-trichloroethane	(f) Half-life of 1,1,1-trichloroethane	C5 Kinetics	P2	Low
2017 Q2	Atmospheric chemistry	(g) Global OH radical concentration	(g) Global OH radical concentration	C3 Stoichiometry	P1	Low
2017 Q3	The chemistry of Iron Man	(a) TiCl4 + NH3 for TiN films	(a) TiCl4 + NH3 for TiN films	C10 Inorganic Chemistry	P3	Low
2017 Q3	The chemistry of Iron Man	(c) Coordination in α-Ti3Au (Au at corners, Ti at face centres; a = 4.15 Å)	(c) Coordination in α-Ti3Au (Au at corners, Ti at face centres; a = 4.15 Å)	C12 Clearly Beyond the Syllabus	P4	Very Low
2017 Q3	The chemistry of Iron Man	(d) α-Ti3Au distances	(d) α-Ti3Au distances	C12 Clearly Beyond the Syllabus	P4	Very Low
2017 Q3	The chemistry of Iron Man	(e) Coordination in β-Ti3Au (Au at corners + body centre; Ti in face pairs; a = 5.09 Å)	(e) Coordination in β-Ti3Au (Au at corners + body centre; Ti in face pairs; a = 5.09 Å)	C12 Clearly Beyond the Syllabus	P4	Very Low
2017 Q3	The chemistry of Iron Man	(f) β-Ti3Au distances	(f) β-Ti3Au distances	C12 Clearly Beyond the Syllabus	P4	Very Low
2017 Q5	Superbases	(e) B2-, C2-, and DEB2-	(e) B2-, C2-, and DEB2-	C10 Inorganic Chemistry	P3	Low
2016 Q2	The chemistry of tungsten	(a) The tungstate ion	(a) The tungstate ion	C10 Inorganic Chemistry	P3	Low
2016 Q2	The chemistry of tungsten	(b) Scheelite to tungsten metal	(b) Scheelite to tungsten metal	C10 Inorganic Chemistry	P3	Low
2016 Q2	The chemistry of tungsten	(f) WF6 + water	(f) WF6 + water	C10 Inorganic Chemistry	P3	Low
2016 Q2	The chemistry of tungsten	(h) Counting atoms in the scheelite unit cell	(h) Counting atoms in the scheelite unit cell	C12 Clearly Beyond the Syllabus	P4	Very Low
2016 Q2	The chemistry of tungsten	(i) Density of scheelite	(i) Density of scheelite	C12 Clearly Beyond the Syllabus	P4	Very Low
2016 Q5	Dating King Richard III	(a) Removing the inorganic content — circle one	(a) Removing the inorganic content — circle one	C11 Experiments & Graphs	P3	Low
2016 Q5	Dating King Richard III	(b) Polyglycine to graphite	(b) Polyglycine to graphite	C3 Stoichiometry	P1	Low

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2016 Q5	Dating King Richard III	(c) 14C decays per day at time of death	(c) 14C decays per day at time of death	C3 Stoichiometry	P1	Low
2016 Q5	Dating King Richard III	(d) Age of the bones and date of death	(d) Age of the bones and date of death	C3 Stoichiometry	P1	Low
2015 Q1	The chemistry of touchscreens	(a) Indium(III) oxide from the hydroxide	(a) Indium(III) oxide from the hydroxide	C10 Inorganic Chemistry	P3	Low
2015 Q1	The chemistry of touchscreens	(c) Indium ions in the FCC cell	(c) Indium ions in the FCC cell	C12 Clearly Beyond the Syllabus	P4	Very Low
2015 Q1	The chemistry of touchscreens	(d) Oxide ions in the cell	(d) Oxide ions in the cell	C12 Clearly Beyond the Syllabus	P4	Very Low
2015 Q1	The chemistry of touchscreens	(e) Product of heating In2O3 in air	(e) Product of heating In2O3 in air	C10 Inorganic Chemistry	P3	Low
2015 Q1	The chemistry of touchscreens	(f) Product with ammonia at 630 °C	(f) Product with ammonia at 630 °C	C10 Inorganic Chemistry	P3	Low
2015 Q2	Detecting molecules in space	(d) Rotational constant of HC11N	(d) Rotational constant of HC11N	C12 Clearly Beyond the Syllabus	P4	Very Low
2015 Q2	Detecting molecules in space	(f) Reduced mass of 12C16O	(f) Reduced mass of 12C16O	C12 Clearly Beyond the Syllabus	P4	Very Low
2015 Q2	Detecting molecules in space	(g) Bond length and transition assignment of CO	(g) Bond length and transition assignment of CO	C12 Clearly Beyond the Syllabus	P4	Very Low
2015 Q4	Hangovers	(h) Sources of the toxic metabolites	(h) Sources of the toxic metabolites	C8 Organic Chemistry	P2	Low
2015 Q4	Hangovers	(i) Metabolism of the poisonous alcohol — tick all that apply	(i) Metabolism of the poisonous alcohol — tick all that apply	C8 Organic Chemistry	P2	Low
2015 Q5	"Green" jet fuel	(b) Fischer–Tropsch	(b) Fischer–Tropsch	C10 Inorganic Chemistry	P3	Low
2015 Q5	"Green" jet fuel	(c) Ceria-mediated reductions	(c) Ceria-mediated reductions	C10 Inorganic Chemistry	P3	Low
2014 Q1	Controlling phosphate levels	(b) Precipitation of lanthanum carbonate	(b) Precipitation of lanthanum carbonate	C10 Inorganic Chemistry	P3	Low
2014 Q1	Controlling phosphate levels	(c) The anhydrous carbonate route	(c) The anhydrous carbonate route	C10 Inorganic Chemistry	P3	Low
2014 Q1	Controlling phosphate levels	(d) From lanthanum chloride and ammonium hydrogencarbonate	(d) From lanthanum chloride and ammonium hydrogencarbonate	C10 Inorganic Chemistry	P3	Low
2014 Q4	Bombardier beetles	(a) Catalase reaction	(a) Catalase reaction	C10 Inorganic Chemistry	P3	Low
2014 Q5	Fire and ice: methane clathrates	(e) The unit cell	(e) The unit cell	C12 Clearly Beyond the Syllabus	P4	Very Low

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2013 Q1	Rocket fuels	(c) C-Stoff / T-Stoff	(c) C-Stoff / T-Stoff	C10 Inorganic Chemistry	P3	Low
2013 Q3	Coloured compounds	(b) The T–Z mixing grid	(b) The T–Z mixing grid	C10 Inorganic Chemistry	P3	Low
2013 Q5	Getting big muscles: creatine	(h) The lipophilic supplement	(h) The lipophilic supplement	C8 Organic Chemistry	P2	Low
2012 Q1	Sulfur chemistry	(c) Orthorhombic vs monoclinic S8	(c) Orthorhombic vs monoclinic S8	C10 Inorganic Chemistry	P3	Low
2012 Q3	Chemistry general knowledge	(a) Order of discovery	(a) Order of discovery	C10 Inorganic Chemistry	P3	Low
2012 Q3	Chemistry general knowledge	(c) Formulas	(c) Formulas	C10 Inorganic Chemistry	P3	Low
2012 Q3	Chemistry general knowledge	(e) Zinc oxide on heating	(e) Zinc oxide on heating	C10 Inorganic Chemistry	P3	Low
2012 Q3	Chemistry general knowledge	(f) Tea chemistry	(f) Tea chemistry	C10 Inorganic Chemistry	P3	Low
2012 Q5	The world's smallest powered car	(e) Motion of each car on copper	(e) Motion of each car on copper	C11 Experiments & Graphs	P3	Low
2011 Q2	Calcium carbide	(c) Formation of calcium carbide	(c) Formation of calcium carbide	C10 Inorganic Chemistry	P3	Low
2011 Q2	Calcium carbide	(d) Formula of hydrocarbon C	(d) Formula of hydrocarbon C	C10 Inorganic Chemistry	P3	Low
2011 Q2	Calcium carbide	(e) Carbide plus water	(e) Carbide plus water	C10 Inorganic Chemistry	P3	Low
2011 Q2	Calcium carbide	(f) Phosphate impurity with coke	(f) Phosphate impurity with coke	C10 Inorganic Chemistry	P3	Low
2011 Q2	Calcium carbide	(g) The unpleasant odour	(g) The unpleasant odour	C10 Inorganic Chemistry	P3	Low
2010 Q3	Halogen fluorides	(a) Uranium + ClF3	(a) Uranium + ClF3	C10 Inorganic Chemistry	P3	Low
2010 Q4	Salting roads in winter	(a) The chemical-potential sketch	(a) The chemical-potential sketch	C12 Clearly Beyond the Syllabus	P4	Very Low
2010 Q6	Gold leaf	(b) Atoms in the unit cell	(b) Atoms in the unit cell	C12 Clearly Beyond the Syllabus	P4	Very Low
2010 Q6	Gold leaf	(c) Cell dimensions in terms of r	(c) Cell dimensions in terms of r	C12 Clearly Beyond the Syllabus	P4	Very Low
2010 Q6	Gold leaf	(d) Molar volume	(d) Molar volume	C12 Clearly Beyond the Syllabus	P4	Very Low
2010 Q6	Gold leaf	(e) Packing fraction	(e) Packing fraction	C12 Clearly Beyond the Syllabus	P4	Very Low
2010 Q6	Gold leaf	(f) Radius of a gold atom	(f) Radius of a gold atom	C12 Clearly Beyond the Syllabus	P4	Very Low
2010 Q6	Gold leaf	(g) The Dome of the Rock	(g) The Dome of the Rock	C3 Stoichiometry	P1	Low
2009 Q1	Environmentally friendly fireworks	(a) Dihydrazinotetrazine	(a) Dihydrazinotetrazine	C10 Inorganic Chemistry	P3	Low

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2009 Q3	The chemistry of matches	(a) The match-head reaction	(a) The match-head reaction	C10 Inorganic Chemistry	P3	Low
2009 Q5	Arsenic	(a) Arsine	(a) Arsine	C10 Inorganic Chemistry	P3	Low
2009 Q5	Arsenic	(b) Generating arsine	(b) Generating arsine	C10 Inorganic Chemistry	P3	Low
2009 Q6	The hidden van Gogh (XRF)	(c) Element A from Kα(A) (≈ 8000 eV)	(c) Element A from Kα(A) (≈ 8000 eV)	C12 Clearly Beyond the Syllabus	P4	Very Low
2009 Q6	The hidden van Gogh (XRF)	(d) Element B from Lα(B) (≈ 10 000 eV)	(d) Element B from Lα(B) (≈ 10 000 eV)	C12 Clearly Beyond the Syllabus	P4	Very Low
2009 Q6	The hidden van Gogh (XRF)	(e) The 10 500 eV peak	(e) The 10 500 eV peak	C12 Clearly Beyond the Syllabus	P4	Very Low
2009 Q6	The hidden van Gogh (XRF)	(f) Kα of antimony	(f) Kα of antimony	C12 Clearly Beyond the Syllabus	P4	Very Low
2008 Q1	Pure silicon for solar cells	(a) Quartz + carbon	(a) Quartz + carbon	C10 Inorganic Chemistry	P3	Low
2008 Q1	Pure silicon for solar cells	(b) Silane and trichlorosilane	(b) Silane and trichlorosilane	C10 Inorganic Chemistry	P3	Low
2008 Q1	Pure silicon for solar cells	(c) The ethanol route	(c) The ethanol route	C10 Inorganic Chemistry	P3	Low
2008 Q1	Pure silicon for solar cells	(d) Zone refining	(d) Zone refining	C11 Experiments & Graphs	P3	Low
2008 Q4	Aluminium chemistry and rat poison	(c) Hydrolysis of AlP	(c) Hydrolysis of AlP	C10 Inorganic Chemistry	P3	Low
2008 Q4	Aluminium chemistry and rat poison	(e) Compound B (m/z max 864)	(e) Compound B (m/z max 864)	C10 Inorganic Chemistry	P3	Low
2008 Q4	Aluminium chemistry and rat poison	(f) Compound C	(f) Compound C	C10 Inorganic Chemistry	P3	Low
2008 Q4	Aluminium chemistry and rat poison	(g) Gas D	(g) Gas D	C10 Inorganic Chemistry	P3	Low
2008 Q5	Toxins from cone snails (mass-spec sequencing)	(a) Ile–Leu–Gly	(a) Ile–Leu–Gly	C9 Structural Analysis	P3	Low
2008 Q5	Toxins from cone snails (mass-spec sequencing)	(b) Number of unique sequences	(b) Number of unique sequences	C9 Structural Analysis	P3	Low
2008 Q5	Toxins from cone snails (mass-spec sequencing)	(c) Sequence of the first 7 amino acids	(c) Sequence of the first 7 amino acids	C9 Structural Analysis	P3	Low
2007 Q2	Geometric shapes in chemistry	(h)–(k) The molybdophosphate ion	(h)–(k) The molybdophosphate ion	C10 Inorganic Chemistry	P3	Low
2007 Q3	Phosphate levels in our environment	(d) Lanthanum carbonate + stomach acid	(d) Lanthanum carbonate + stomach acid	C10 Inorganic Chemistry	P3	Low
2006 Q2	Redox chemistry of vehicle pollution	(d) The ceria oxygen-storage equilibrium	(d) The ceria oxygen-storage equilibrium	C10 Inorganic Chemistry	P3	Low
2006 Q2	Redox chemistry of vehicle pollution	(g) Urea + water → ammonia	(g) Urea + water → ammonia	C10 Inorganic Chemistry	P3	Low
2006 Q2	Redox chemistry of vehicle pollution	(h) NH3 with NO and NO2	(h) NH3 with NO and NO2	C10 Inorganic Chemistry	P3	Low

Year & Question	Question Title	Sub-question	Topic	Module	Tier	Interview Frequency
2006 Q3	Acyl chlorides and related oxochlorides	(e) Thionyl chloride + water	(e) Thionyl chloride + water	C10 Inorganic Chemistry	P3	Low
2006 Q4	Thermal decomposition of copper(II) sulfate	(a) Formulae of A, B and C	(a) Formulae of A, B and C	C10 Inorganic Chemistry	P3	Low
2006 Q5	Emergency oxygen supplies	(a) Decomposition of sodium chlorate(V)	(a) Decomposition of sodium chlorate(V)	C10 Inorganic Chemistry	P3	Low
2006 Q5	Emergency oxygen supplies	(f) Potassium superoxide chemistry	(f) Potassium superoxide chemistry	C10 Inorganic Chemistry	P3	Low
2005 Q1	Carbon oxides	(c) PdCl2 test for CO	(c) PdCl2 test for CO	C10 Inorganic Chemistry	P3	Low
2005 Q2	Diiodine pentoxide	(a) I2O5 + CO	(a) I2O5 + CO	C10 Inorganic Chemistry	P3	Low
2005 Q2	Diiodine pentoxide	(c) The hydrated form	(c) The hydrated form	C10 Inorganic Chemistry	P3	Low
2005 Q2	Diiodine pentoxide	(d) Parent acid of I2O5	(d) Parent acid of I2O5	C10 Inorganic Chemistry	P3	Low
2005 Q2	Diiodine pentoxide	(f) I2 + Cl2 + H2O	(f) I2 + Cl2 + H2O	C10 Inorganic Chemistry	P3	Low
2005 Q3	Ants and methanoic acid	(a)(ii) Ants per litre	(a)(ii) Ants per litre	C3 Stoichiometry	P1	Low
2005 Q5	The Breathalyser	(a)(iv) With potassium and sulfuric acid	(a)(iv) With potassium and sulfuric acid	C10 Inorganic Chemistry	P3	Low
2005 Q5	The Breathalyser	(b)(iii) Colour change	(b)(iii) Colour change	C11 Experiments & Graphs	P3	Low
2004 Q2	Pollution and the Taj Mahal	(d)(i) Ba(OH)2 + CO2	(d)(i) Ba(OH)2 + CO2	C10 Inorganic Chemistry	P3	Low
2004 Q3	Dissolved oxygen (the Winkler method)	(b) Indicator	(b) Indicator	C11 Experiments & Graphs	P3	Low
2004 Q3	Dissolved oxygen (the Winkler method)	(d) The colourless gas	(d) The colourless gas	C10 Inorganic Chemistry	P3	Low
2004 Q5	Tin pest	(g) Greater coordination number	(g) Greater coordination number	C10 Inorganic Chemistry	P3	Low
2003 Q1	Heating a cup of coffee	(a) CaO + water	(a) CaO + water	C10 Inorganic Chemistry	P3	Low
2003 Q2	Reinecke's salt, NH4[Cr(SCN)x(NH3)γ]	(a) x and y	(a) x and y	C12 Clearly Beyond the Syllabus	P4	Very Low
2003 Q6	Hydroxylamine and iron(III)	(d) The nitrogen product	(d) The nitrogen product	C10 Inorganic Chemistry	P3	Low
2003 Q8	Polonium	(b) Alpha-decay product of 210Po	(b) Alpha-decay product of 210Po	C12 Clearly Beyond the Syllabus	P4	Low
2003 Q8	Polonium	(c) Power output after 1 year	(c) Power output after 1 year	C3 Stoichiometry	P1	Low
2003 Q8	Polonium	(d) Half-life of 238Pu	(d) Half-life of 238Pu	C5 Kinetics	P2	Low
2003 Q8	Polonium	(e) Atomic radius of polonium	(e) Atomic radius of polonium	C1 Atomic Structure & Periodicity	P2	Low